



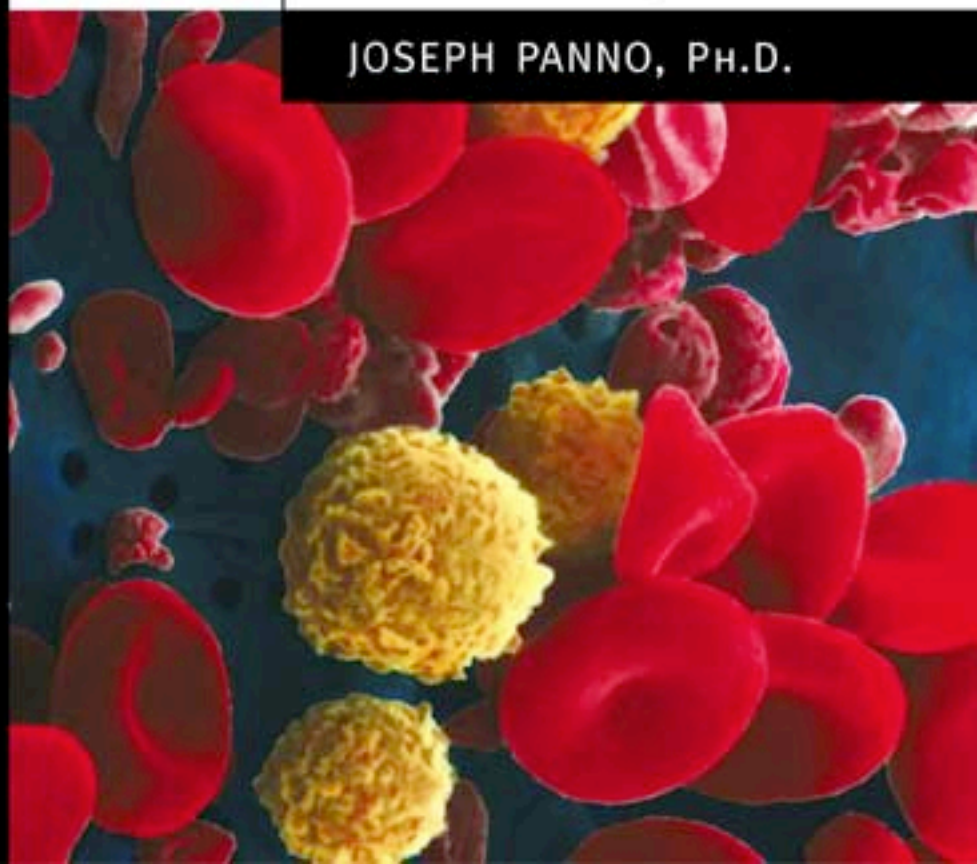
THE
newbiology

THE CELL



*Evolution of the
First Organism*

JOSEPH PANNO, PH.D.



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Evolution of the First Organism

Joseph Panno, Ph.D.



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THE CELL: Evolution of the First Organism

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This book is printed on acid-free paper.

*For my wife, Diana,
who worked with me in the lab for many years,
and for my daughter Eleanor,
who knew about cells before she could read or write.*



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PREFACE



The New Biology set consists of the following six volumes: *The Cell*, *Animal Cloning*, *Stem Cell Research*, *Gene Therapy*, *Cancer*, and *Aging*. The set is intended primarily for middle and high school students, but it is also appropriate for first-year university students and the general public. In writing this set, I have tried to balance the need for a comprehensive presentation of the material, covering many complex fields, against the danger of burying—and thereby losing—young students under a mountain of detail. Thus the use of lengthy discussions and professional jargon has been kept to a minimum, and every attempt has been made to ensure that this be done without sacrificing the important elements of each topic. A large number of drawings are provided throughout the series to illustrate the subject matter.

The term *new biology* was coined in the 1970s with the introduction of recombinant DNA technology (or biotechnology). At that time, biology was largely a descriptive science in danger of going adrift. Microbiologists at the turn of the century had found cures for a few diseases, and biologists in the 1960s had cracked the genetic code, but there was still no way to study the function of a gene or the cell as a whole. Biotechnology changed all that, and scientists of the period referred to it as the new technique or the new biology. However, since that time it has become clear that the advent of biotechnology was only the first step toward a new biology, a biology that now includes nuclear transfer technology (animal cloning), gene therapy, and stem cell therapy. All these technologies are covered in the six volumes of this set.

The cell is at the very heart of the new biology and thus figures prominently in this book series. Biotechnology was specifically designed for studying cells, and using those techniques, scientists gained insights into cell structure and function that came with unprecedented detail.

As knowledge of the cell grew, the second wave of technologies—animal cloning, stem cell therapy, and gene therapy—began to appear throughout the 1980s and 1990s. The technologies and therapies of the new biology are now being used to treat a wide variety of medical disorders, and someday they may be used to repair a damaged heart, a severed spinal cord, and perhaps even reverse the aging process. These procedures are also being used to enhance food crops and the physical characteristics of dairy cows and to create genetically modified sheep that produce important pharmaceuticals. The last application alone could save millions of lives every year.

While the technologies of the new biology have produced some wonderful results, some of the procedures are very controversial. The ability to clone an animal or genetically engineer a plant raises a host of ethical questions and environmental concerns. Is a cloned animal a freak that we are creating for our entertainment, or is there a valid medical reason for producing such animals? Should we clone ourselves, or use the technology to re-create a loved one? Is the use of human embryonic stem cells to save a patient dying from leukemia a form of high-tech cannibalism? These and many other questions are discussed throughout the series.

The New Biology set is laid out in a specific order, indicated previously, that reflects the natural progression of the discipline. That is, knowledge of the cell came first, followed by animal cloning, stem cell therapy, and gene therapy. These technologies were then used to expand our knowledge of, and develop therapies for, cancer and aging. Although it is recommended that *The Cell* be read first, this is not essential. Volumes 2 through 6 contain extensive background material, located in the final chapter, on the cell and other new biology topics. Consequently, the reader may read the set in the order he or she prefers.

ACKNOWLEDGMENTS



I would first like to thank my friend and mentor, the late Dr. Karun Nair, for helping me understand some of the intricacies of the biological world and for encouraging me to seek that knowledge by looking beyond the narrow confines of any one discipline. The clarity and accuracy of the initial manuscript for this book was greatly improved by reviews and comments from Diana Dowsley and Michael Panno, and later by Frank Darmstadt, Executive Editor; Dorothy Cummings, Project Editor; and Anthony Sacramone, Copy Editor. I am also indebted to Ray Spangenburg, Kit Moser, Sharon O'Brien, and Diana Dowsley for their help in locating photographs for the New Biology set. Finally, I would like to thank my wife and daughter, to whom this book is dedicated, for the support and encouragement that all writers need and are eternally grateful for.

INTRODUCTION



Life began in the oceans of ancient Earth more than 3 billion years ago. At that time, our planet was a wild and stormy place with an atmosphere that was not fit to breathe. Although the storms were violent, life could not have appeared without them. The lightning provided the energy to make certain molecules that all living things need, and the wind churned up the surface of the soupy seas like a diligent cook stirring a pot. The storms had to stir that pot for a billion years before it finally happened: a tiny bubble, too small to see with the naked eye, gave birth to the first cell, and in the wink of a cosmic eye, the earth was teeming with life.

The first cells were little more than microscopic bags of chemicals that were capable of reproduction. They lived solitary lives but eventually, after learning how to communicate with each other, began to form small colonies consisting of no more than three or four cells each. As time passed, cell-to-cell communication and cooperation became so elaborate that the first simple colonies were transformed into complex multicellular plants and animals. The first cell colonies were produced almost 3 billion years ago by prokaryotes, a simple kind of cell more commonly known as bacteria. Supercells called eukaryotes, appearing about 2 billion years ago, created much more elaborate colonies that eventually gave rise to true multicellular organisms. Eukaryotes are the direct descendants of the prokaryotes, but they are larger, more complex, and more adept at cell-to-cell communication. Plants and animals are all made from eukaryotes. The human body, for example, is made from more than 100 billion eukaryotes, a population consisting of more than 200 different cell types that are organized into organs and tissues. Our brains alone are constructed from 10 billion eukaryotes called neurons that are linked together in a network of enormous complexity. Some neurons in our brain are capable of communicating simultane-

ously with 100,000 other neurons. It is this level of complexity that produces our intellect and gives us the powers of speech and vision. In some sense, the human brain is the ultimate colony, the most intricate cellular community ever to appear on Earth.

Our understanding of the cell has increased tremendously since the 1970s, when recombinant DNA technology was first introduced. This technology made it possible to study the structure and function of a cell in minute detail. Prior to the 1970s, biologists had only a basic understanding of the cell; they knew the DNA was located in the nucleus, that the cell was surrounded by what appeared to be a featureless membrane, and that the cell interior was full of structures called organelles, but their functions were largely unknown. Today scientists have sequenced the entire human genome, as well as the genomes of many other organisms. They have determined the function of virtually every cellular organelle, and they have shown that the cell membrane, far from being featureless, contains a molecular forest that gives the cell its eyes, its ears, and the equipment it needs to capture food and to communicate with other cells.

By studying the cell, we improve our understanding of the living world and, in particular, our understanding of plant and animal physiology, genetics, and biochemistry. This wealth of information has revolutionized the biological and medical sciences. For human society, this knowledge translates into a dramatic reduction in mortalities due to infectious diseases and medical disorders. The war on cancer, begun more than 20 years ago, is finally approaching a stage where all cancers will be curable. Improved treatment and prognosis is now possible for many other disorders, such as cardiovascular disease, diabetes, and cystic fibrosis. Improved knowledge of the cell made it possible for researchers to isolate and culture stem cells, a very resourceful kind of cell that may be used to treat spinal cord trauma and degenerative neurological diseases such as Alzheimer's disease and Parkinson's disease. In 1996, scientists in Scotland, using newly acquired knowledge about the way cells divide, performed the most dramatic biological experiment ever conducted when they cloned a sheep named Dolly. Far from being a clever trick, this accomplishment may provide the world with a ready source of therapeutic drugs that could save millions of human lives each year.

This book, the first in the New Biology series, covers the structure and function of the cell with a special emphasis on cell division and cell-to-cell communication (also known as cell signaling). The cell's ability to communicate was essential for the development of multicellular creatures. Moreover, the corruption of that ability, and the process of cell division, is central to many pathological conditions such as cancer and Alzheimer's disease. The corruption of cell signaling and cell division are also responsible for many of the changes that occur in animals as they grow old. The first three chapters of this book discuss the origin of life, the emergence of prokaryotes, and the appearance of eukaryotes. Eukaryotes are the main subject of the book. Subsequent chapters are devoted to discussions of the cell cycle, genes and genetic mechanisms, and the transition, initiated by eukaryotes, from single cells to multicellular organisms. One chapter is devoted to the neuron, an especially talented eukaryote that made the transition to multicellular creatures a possibility. The final chapter provides background material on recombinant DNA technology and other topics that are relevant to cell biology.

THE ORIGIN OF LIFE

Life began so long ago that many people believe it is impossible to reconstruct the events that led to the appearance of the first cell. The skepticism is understandable, since there are no fossils from that period to study and our knowledge of the Earth's formative years is still rudimentary. Nevertheless, some progress has been made by studying the most primitive cells on Earth today and by conducting laboratory experiments that attempt to reconstruct, in a test tube, the conditions of ancient Earth.

The Big Bang

Fifteen billion years ago, everything in the universe was a soupy concoction of plasma compressed into an area smaller than the head of a pin. There was no matter as we think of it now: no iron, no copper, no carbon, and no oxygen. Just subatomic particles brought together by a crushing force of gravity. No one knows how long the universe remained in this state or even if time, as we know it, existed. We do know that it was extremely hot, with temperatures exceeding 10 billion degrees, 1,000 times hotter than the center of the Sun. Eventually, something happened (no one knows what), and that pinhead of unimaginable heat and density suddenly exploded. Within seconds, the temperature dropped enough for atomic nuclei to form; after a million years, the temperature was low enough for the first elements to appear. The first of these was hydrogen, the simplest of all elements, and the one that gave rise to all the rest. Although the universe was cooling down, it was still hot enough to fuse hydrogen atoms to produce helium. Enough

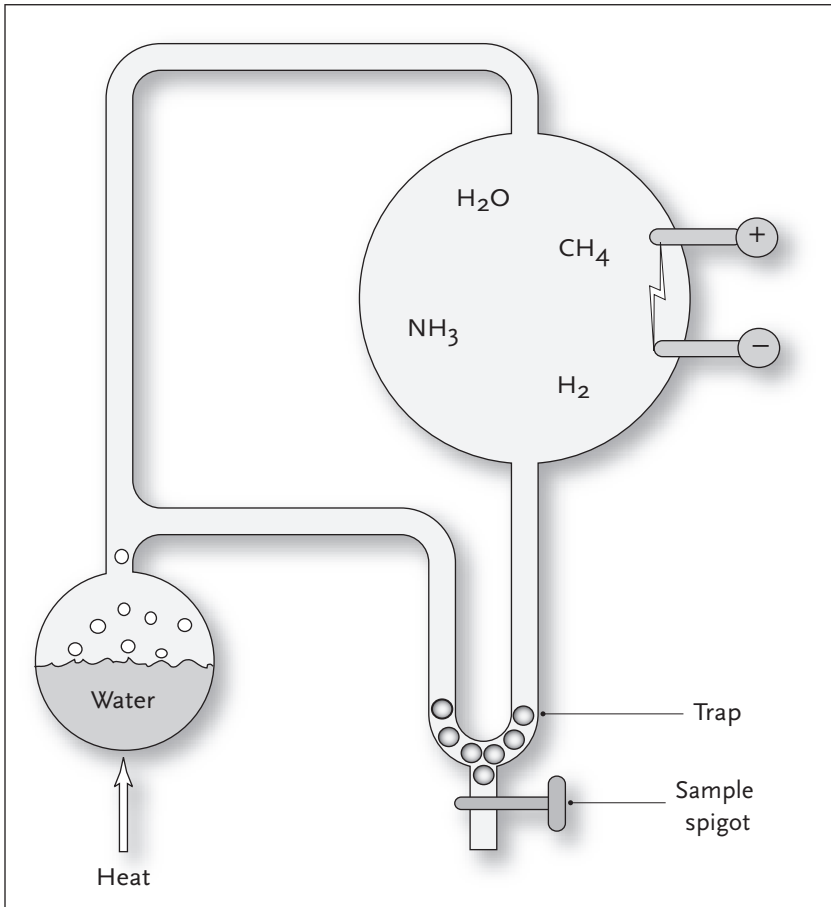
hydrogen and helium were formed in this way to produce all the stars and galaxies. Heat within the stars was sufficient to fuse hydrogen and helium atoms to form all the other elements that we now find in nature, such as carbon, iron, copper, and nitrogen.

The Importance of Violent Storms

Ten billion years after the big bang, our Earth was created as a molten ball of metal and stone thrown off by the sun during the formation of the solar system. Additional material was added to our planet as it collided with asteroids and meteors. The high surface temperature liberated an enormous amount of water vapor from the nearly molten rocks. The vapor rose into the atmosphere, forming a heavy cloud layer that completely enshrouded the planet, effectively blocking the sun's rays. During the subsequent half-billion years, the Earth cooled down, and when it did, the rains began to fall. This was no brief summer shower, but a pelting rain that lasted hundreds of years and led to the formation of the oceans, which covered most of the Earth's surface just as they do



An artist's conception of prebiotic Earth showing a volcano and the hot, stormy environment. (Courtesy of Steve Munsinger/Photo Researchers, Inc.)



Urey-Miller experiment to simulate conditions on prebiotic Earth. Water is heated in a closed system containing methane (CH_4), ammonia (NH_3), and hydrogen (H_2) gases. An electric discharge is passed through the vaporized mixture to simulate lightning in the atmosphere. Synthesized compounds collect in the trap and are sampled by opening the spigot. The original experiment was run for a week or more before samples were collected.

today. The land was barren and wracked with volcanic eruptions that spewed noxious gases such as methane (CH_4) and ammonia (NH_3) into the atmosphere. The air contained very little, if any, free oxygen.

A planet with an atmosphere of methane and ammonia does not, at first glance, appear a likely candidate for the origin of life. Modern cells

need oxygen to breathe and require four kinds of organic molecules: amino acids (building blocks for proteins), nucleic acids (building blocks for DNA and RNA), fats, and sugars. This is a short list, but a long way from methane and ammonia. Nevertheless, in 1953, Harold Urey, a professor at the University of Chicago, and his graduate student, Stanley Miller, decided to test the hypothesis that Earth's ancient atmosphere, combined with fierce electric storms, was essential for the production of the molecules that cells need to live.

To conduct the experiment, Miller constructed a simple test-tube apparatus consisting of two round flasks connected by glass tubing. One of the flasks, containing water, simulated the ocean; a second flask, filled with hydrogen, methane, and ammonia gases, served as the atmosphere. They passed an electric discharge through the flask containing the atmosphere to simulate lightning and heated the water flask to produce the high temperature of the young earth. After a week, Urey and Miller tested the contents of the flask and to their great surprise found that the water contained large amounts of amino acids. By varying the conditions of their experiment they were able to produce a wide variety of organic compounds, including nucleic acids, sugars, and fats.

Essential Molecules Formed Spontaneously

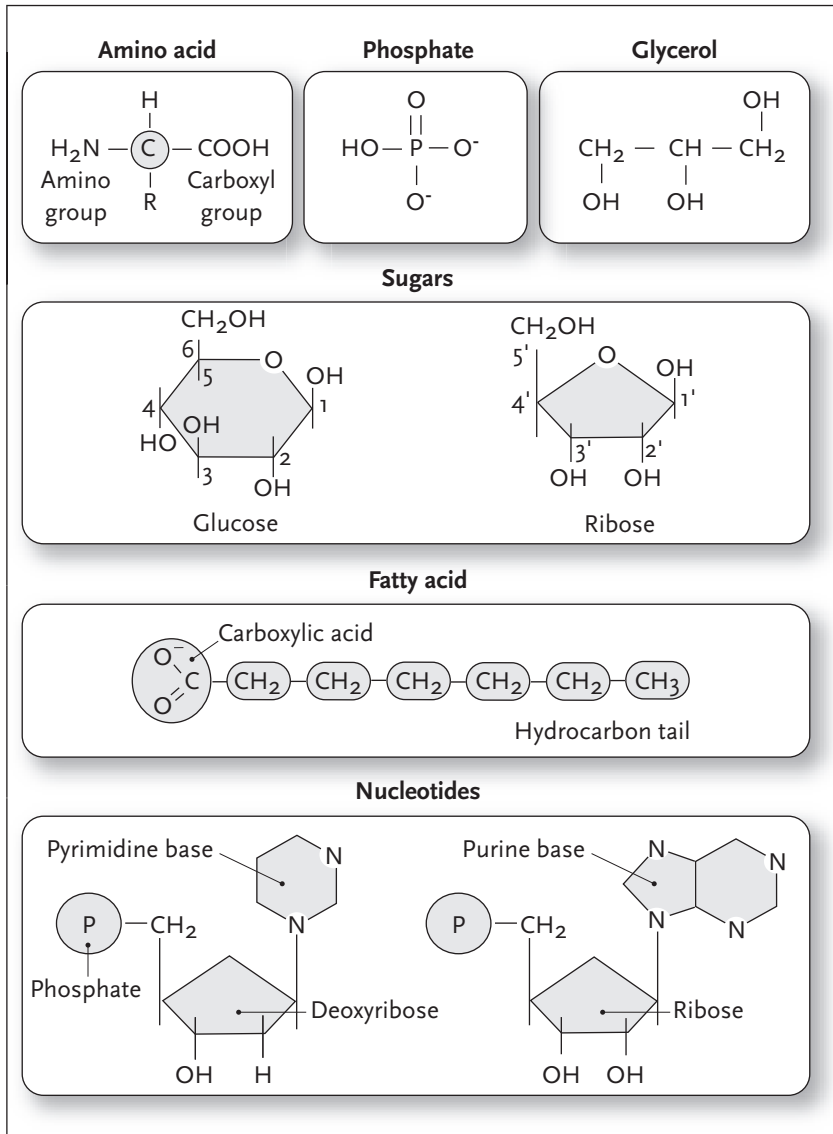
The Urey-Miller experiment made it clear that the basic building blocks for life could have been made in the harsh, prebiotic (before life) Earth environment. Given the conditions of that period, it now seems almost inevitable that such molecules would be synthesized. These results, published in 1953, generated a great deal of excitement, both in the science community and among the general public. Many people believed we were close to understanding the origin of life itself, a feat that had seemed impossible just a few years earlier. The Urey-Miller experiment suggested that the conditions on earth 4.5 billion years ago led to the formation of certain key organic molecules, which assembled themselves into larger molecules that eventually went on to form the first cell. However, much of the optimism that this experiment generated began to fade under the cloud of an impenetrable paradox.

Modern cells depend heavily on an interaction between proteins and nucleic acids (DNA and RNA). The proteins are used to construct

the cell, and a special group of them, called enzymes, control the many chemical reactions that are necessary for cells to live. DNA is a collection of blueprints, or genes, that store the information to make the proteins. One kind of RNA, called messenger RNA (mRNA), serves as an intermediary between the genes and the cell's machinery for synthesizing proteins. Although it is possible for nucleic acids and proteins to self-assemble, it is extremely unlikely that the modern relationship between the three developed spontaneously. It comes down to the age-old question of which came first, the chicken or the egg: the nucleic acids or the proteins? Could DNA, or RNA, have self-assembled and then orchestrated the synthesis of the proteins? Or did the proteins self-assemble and then make their own blueprints using DNA, while ignoring RNA altogether? Before we attempt to resolve this paradox, we must consider in a little more detail the kinds of molecules that cells need to survive.

Modern cells are biochemical entities that synthesize many thousands of molecules. Studying these chemicals, and the biochemistry of the cell, would be a daunting task were it not for the fact that most of the chemical variation is based on six types of molecules that are assembled into just four types of macromolecules. The six basic molecules are amino acids, phosphate, glycerol, sugars, fatty acids, and nucleotides. Amino acids have a simple core structure consisting of an amino group, a carboxyl group, and a variable R group attached to a carbon atom. There are 20 different kinds of amino acids, each with a unique R group. Phosphates are extremely important molecules that are used in the construction, or modification, of many other molecules. They are also used to store chemical-bond energy. Glycerol is a simple three-carbon alcohol that is an important component of cell membranes and fat reservoirs. Sugars are extremely versatile molecules that are used as an energy source and for structural purposes. Glucose, a six-carbon sugar, is the primary energy source for most cells, and it is the principle sugar used to glycosylate proteins and lipids that form the outer coat of all cells. Plants have exploited the structural potential of sugars in their production of cellulose, and thus wood, bark, grasses, and reeds are polymers of glucose and other monosaccharides. Ribose, a five-carbon sugar, is a component of nucleic acids, as well as the cell's main energy depot, adenosine triphosphate (ATP). The numbering convention for sugar carbon atoms is shown in the accompanying

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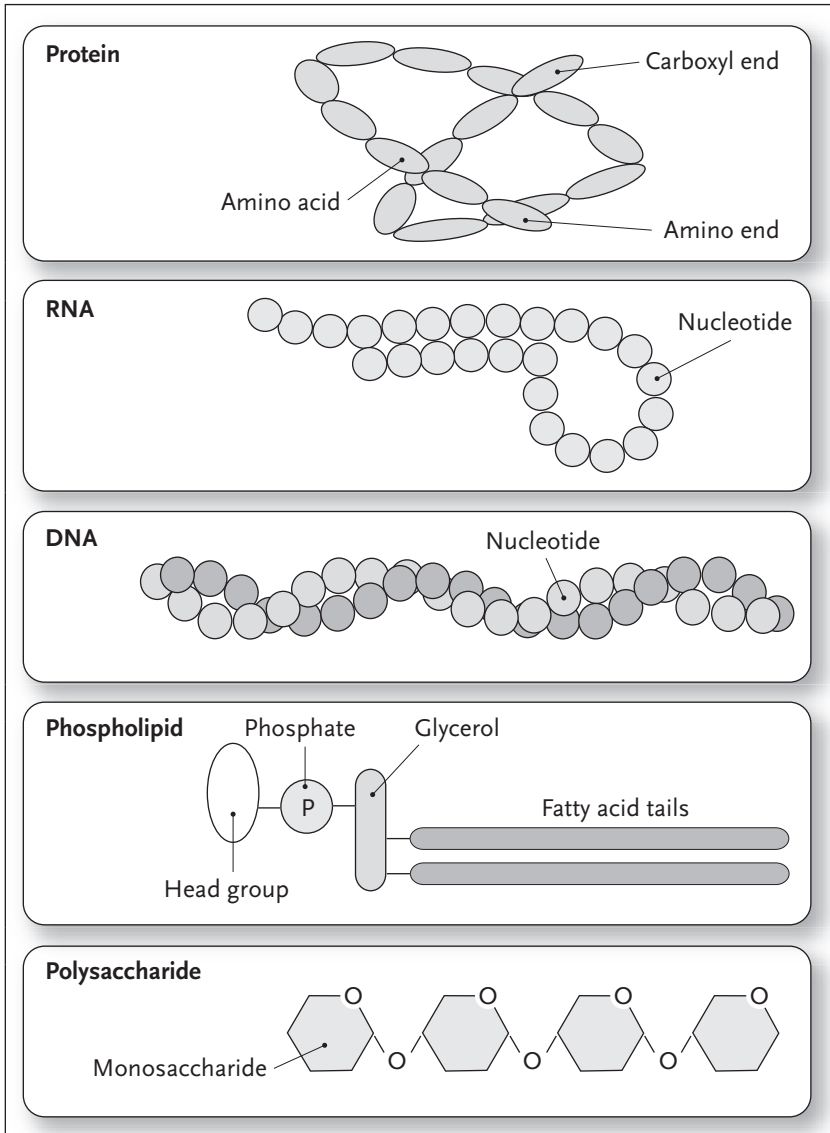


Molecules of the cell. Amino acids are the building blocks of proteins. Phosphate is an important component of many other molecules and is added to proteins to modify their behavior. Glycerol is a three-carbon alcohol that is an important ingredient in cell membranes and fat. Sugars, like glucose, are a primary energy source for most cells and also have many structural functions. Fatty acids are involved in the production of cell membranes and storage of fat. Nucleotides are the building blocks for DNA and RNA.

figure. Ribose carbons are numbered as 1' (1 prime), 2', and so on. Consequently, references to nucleic acids, which include ribose, often refer to the 3' or 5' carbon. Fatty acids consist of a carboxyl group (when ionized it becomes a carboxylic acid) linked to a hydrophobic hydrocarbon tail. These molecules are used in the construction of cell membranes and fat. Nucleotides are building blocks for DNA (deoxyribonucleic acid) and RNA (ribonucleic acid). Nucleotides consist of three components: a phosphate, a ribose sugar, and a nitrogenous (nitrogen containing) ring compound that behaves as a base in solution. Nucleotide bases appear in two forms: a single-ring nitrogenous base, called a pyrimidine, and a double-ringed base, called a purine. There are two kinds of purines (adenine and guanine), and three pyrimidines (uracil, cytosine, and thymine). Uracil is specific to RNA, substituting for thymine. In addition, RNA nucleotides contain ribose, whereas DNA nucleotides contain deoxyribose (hence the origin of their names). Ribose has a hydroxyl (OH) group attached to both the 2' and 3' carbons, whereas deoxyribose is missing the 2' hydroxyl group.

The six basic molecules are used by all cells to construct five essential macromolecules. These include proteins, RNA, DNA, phospholipids, and sugar polymers, known as polysaccharides. Amino acids are linked together by peptide bonds to construct a protein. A peptide bond is formed by linking the carboxyl end of one amino acid to the amino end of a second amino acid. Thus, once constructed, every protein has an amino end, and a carboxyl end. An average protein may consist of 300 to 400 amino acids. Nucleic acids are macromolecules constructed from nucleotides. RNA nucleotides are adenine, uracil, cytosine, and guanine. This nucleic acid is generally single-stranded but can form localized double-stranded regions. RNA is involved in the synthesis of proteins and is a structural and enzymatic component of ribosomes. DNA, a double-stranded nucleic acid, encodes cellular genes and is constructed from adenine, thymine, cytosine, and guanine deoxyribonucleotides (hence, the name deoxyribonucleic acid). The two DNA strands coil around each other like strands in a piece of rope, and for this reason the molecule is known as the double helix. Phospholipids, the main component in cell membranes, are composed of a polar head group (usually an alcohol), a phosphate, glycerol, and two hydrophobic fatty acid tails. Fat that is stored in the body as an energy reserve has a structure similar to

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Macromolecules of the cell. Protein is made from amino acids linked together to form a long chain that can fold up into a three-dimensional structure. RNA and DNA are long chains of nucleotides. RNA is generally single-stranded but can form localized double-stranded regions. DNA is a double-stranded helix, with one strand coiling around the other. A phospholipid is composed of a hydrophilic head-group, a phosphate, a glycerol molecule, and two hydrophobic fatty acid tails. Polysaccharides are sugar polymers.

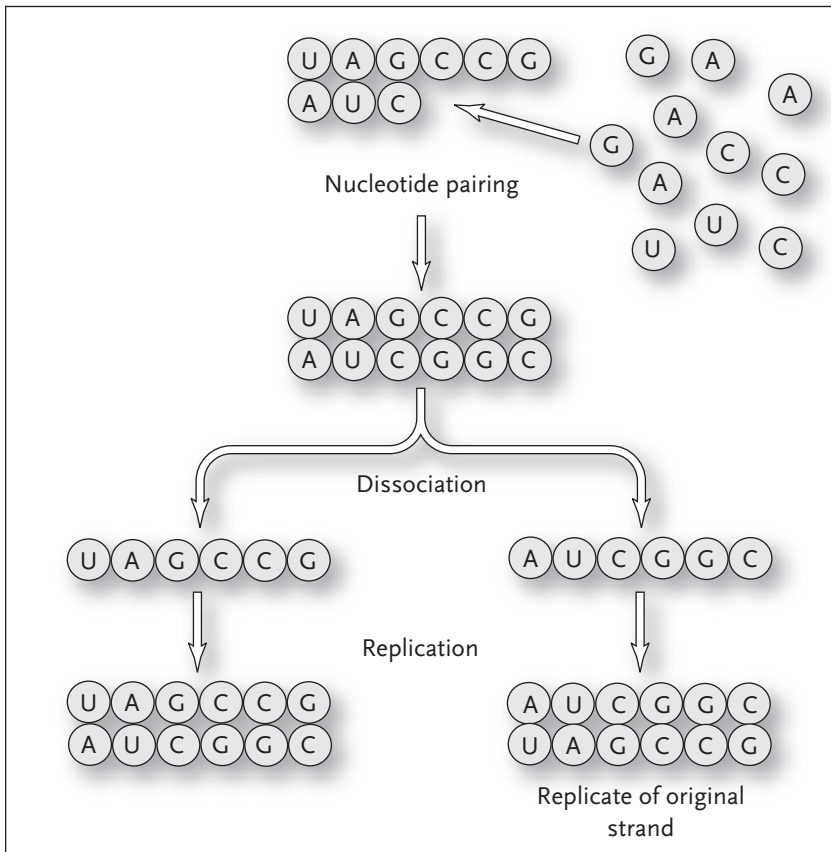
a phospholipid, being composed of three fatty acid chains attached to a molecule of glycerol. The third fatty acid takes the place of the phosphate and head group of a phospholipid. Sugars are polymerized to form chains of two or more monosaccharides. Disaccharides (two monosaccharides), and oligosaccharides (about 3–12 monosaccharides), are attached to proteins and lipids destined for the cell surface. Polysaccharides, such as glycogen and starch, may contain several hundred monosaccharides and are stored in cells as an energy reserve.

All the molecules shown in the accompanying figure are assumed to have formed in the prebiotic oceans, and this was followed by auto-assembly of the macromolecules. Auto-assembly of the nucleic acids could have produced polymers that were 60–100 nucleotides long. With one DNA or RNA strand made, a second strand would have formed automatically through base pairing; that is, the formation of chemical bonds between the nitrogenous bases. The chemistry of these bases is such that cytosine always pairs with guanine, while adenine always pairs with thymine. In the case of RNA, which lacks thymine, adenine pairs with uracil. Thus, pairing is always between a purine and a pyrimidine, and the association between the two can form spontaneously. Base pairing, also known as hybridization, can form between two DNA molecules or between a DNA and an RNA molecule. Self-hybridization, involving a single DNA or RNA molecule, can also occur. Because the early oceans were hot, double-stranded DNA or RNA came apart through dissociation of the two chains. That is, the prevailing heat broke (or melted) the chemical bonds holding each nucleotide pair together without disrupting the two chains or strands. When the strands separate, the cycle repeats with another round of base pairing leading to the production of two more double-stranded molecules, one of which contains the original strand and the other contains its exact copy.

By exploiting the properties of nucleotide base-pairing, coupled with the high temperatures of primitive Earth, short pieces of DNA and RNA can replicate without the aid of any other molecules. In modern cells, DNA remains double-stranded, and in prebiotic earth, with much higher temperatures than we have today, it may still have been slower to dissociate than RNA. Thus RNA replication would have proceeded much more quickly, producing a larger, more diverse population of molecules.

Life Began in an RNA World

The molecule that led to the first living cell would have to be able to replicate itself as well as function as an enzyme. DNA fulfills the first condition, but it has no known enzymatic activity, and at the time of the Urey-Miller experiment, this was thought to be the case for RNA as well: Both were believed to be incapable of regulating chemical reactions, so that neither could build protein molecules by themselves. Pro-



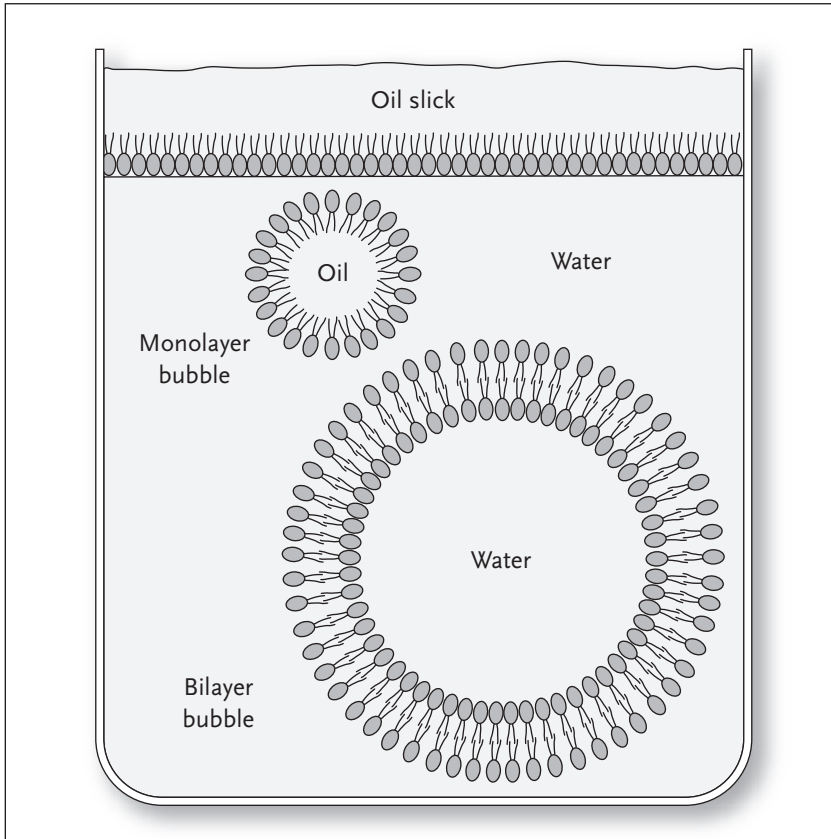
Self-replication of RNA through a process of base pairing and dissociation. Soon after replication, the two double strands separate into four single-stranded molecules, one of which is identical to the original strand. DNA can self-replicate through the same mechanisms.

teins, on the other hand, make efficient enzymes but cannot replicate themselves. This paradox was about to bring an end to origin-of-life studies when, in 1983, Thomas Cech at the University of Colorado, and Sidney Altman at Yale University discovered the ribozyme, an RNA molecule capable of enzymatic activity. This discovery led almost immediately to the suggestion that the first cells came to life in an RNA world.

Ribozymes, assembled in the prebiotic oceans, could not only replicate themselves but also could have catalyzed the formation of specific proteins, which in turn could have functioned as structural proteins or enzymes. Eventually, a protein enzyme appeared that could copy RNA into DNA (such an enzyme, called reverse transcriptase, does exist), and when that happened, the cell's machinery approached a modern level of organization: DNA serving as the blueprint, and RNA acting as an intermediary in the process of protein synthesis. Shifting to a DNA-based genome meant that cells could become much more complex because DNA, as a double-stranded molecule, is more stable than RNA and thus capable of storing information for many more genes.

Once upon a Wave

The concept of the RNA world is very compelling, yet in itself it cannot explain the appearance of the first cell. The auto-assembly of ribozymes and proteins is of little use if they are not confined in some way. But how was this to happen? Organic molecules, newly synthesized by the raging storms, were swept along and dispersed by the wind and currents. If a ribozyme appeared that could make an especially useful protein, the association between the two would have been quickly lost. However, winds sweeping across the ocean have a way of driving things onto shore, so it is possible that organic molecules collected, and were concentrated, along the seashore much like driftwood collecting on a beach. The water near the shore, being shallower than in the open oceans, would also tend to be warmer, concentrating the organic molecules even further through evaporation. Shorelines have another important property that is of interest here. Anyone who has stood on a beach and watched a wave break has witnessed one of the most important mechanisms for the formation of life on this planet:



Phospholipid bubbles. Phospholipid molecules have a hydrophilic headend (shaded ovals) and two hydrophobic tails that do not mix with water and will avoid being surrounded by it. In an oil slick, the hydrophobic tails mix with the oil while the heads stay close to the water. In turbulence, phospholipids form two kinds of bubbles: a monolayer that can only capture a drop of oil and a bilayer that can capture a drop of water. The bilayer allows the hydrophobic tails to associate with themselves, while the heads associate with water on both the inside and outside surfaces of the bubble.

The foam that rolls into shore after the wave breaks is composed of billions of bubbles.

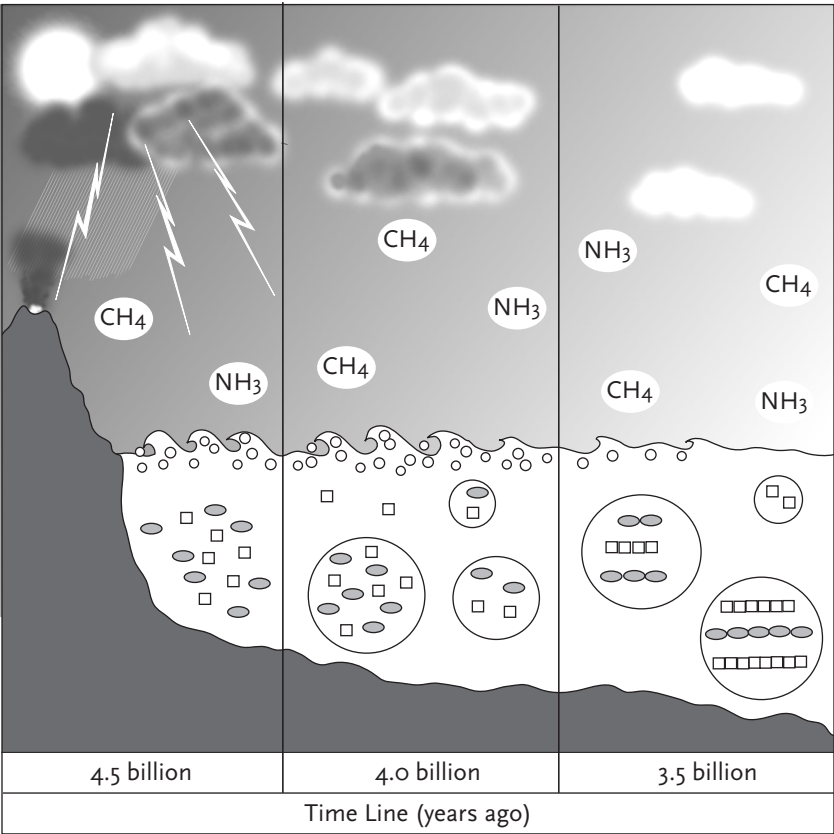
In the prebiotic coastal waters, each bubble that formed collected a different sample of the water and, therefore, represented a unique individual, a separate experiment that could be acted upon by the forces of

natural selection. But as we stand on a beach watching the waves break on the shore, we notice that the foam in the surf disappears very quickly. There is nothing to hold the bubbles together; that is, unless there happens to be a layer of oil on the surface of the water. Bubbles made from oil tend to have a much longer life span. Coincidentally, among the organic molecules synthesized in the prebiotic oceans was the oily compound phospholipid. These molecules may be drawn as a bead, representing the hydrophilic head group, linked to the hydrophobic fatty acid tails. Phospholipids have the unusual property of being hydrophilic (able to mix with water) at the beaded end, but hydrophobic (unable to mix with water) at the tail end. This is curious behavior, but extremely important for the origin of life. Biologists believe that phospholipids were produced by the storms of ancient earth, forming Earth's first oil slick very close to shore, in relatively calm bays and lagoons. Stir up the water with a driving wind and rolling surf, and the phospholipids will produce billions of tiny, stable bubbles.

Phospholipids have another curious, but very important, property: They can form bubbles out of a monolayer (single layer) of molecules or they can form bubbles out of a bilayer (two layers) of molecules. In a monolayer, the external surface of the bubble is always the hydrophilic end of the molecule, whereas the inside of the bubble contains the hydrophobic portion. This type of bubble can only trap oils, not water, and therefore could never lead to the production of a cell. On the other hand, a bubble formed from a bilayer has a hydrophilic surface on both the exterior and interior surfaces. Such a bubble can trap water and water-soluble molecules like ribozymes, sugars, and proteins.

The lipid bilayer is a humble structure, and at first glance may seem as though it is of little consequence, but life could not have arisen without it. Lipid-bilayer bubbles, forming in the seas of ancient Earth, could remain intact long enough to experiment with the molecules and macromolecules they captured when they were formed. If a bubble happened to pick up, or assemble, a protein that stabilized the walls of the bubble, then that bubble had an advantage over the others and would have extra time to experiment with the synthesis of novel ribozymes and proteins. We can imagine this process leading to a primitive form of genetic inheritance. When the bubbles burst, they released the results of their experiments into the water. When new bubbles

formed, they may have captured some, or all, of those molecules and thus have been given a head start through the inheritance of a simple gene pool. This process, acted upon by natural selection, is believed to have transformed the prebiotic bubbles into the first cells.

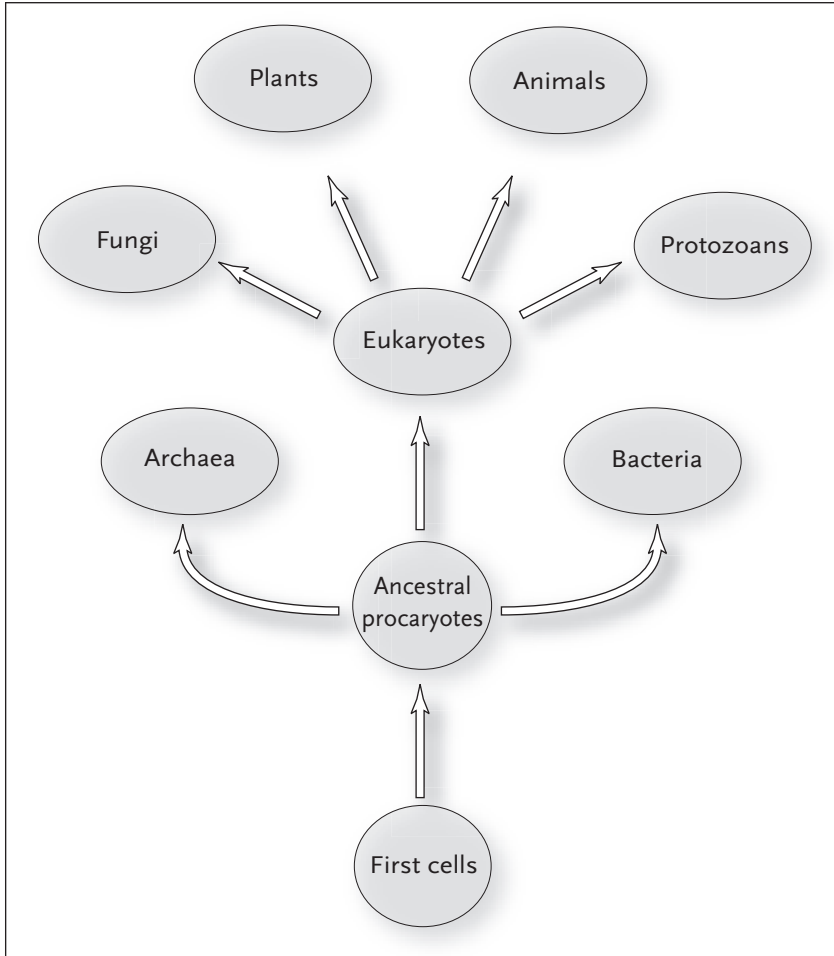


The origin of the first cells. Organic molecules essential for life were synthesized spontaneously 4.5 billion years ago when Earth was hot, stormy, and wracked with constant volcanic eruptions. Some of the organic molecules were captured by lipid bubbles (white circles) formed by ocean turbulence near a shoreline, and by 3.5 million years ago the first cells learned how to assemble the molecules into a variety of polymers. Nucleic acids, amino acids, fats, and sugars were among the organic molecules produced in the prebiotic oceans; only the nucleic acids (white squares) and amino acids (gray ovals) are shown. Major gases in the atmosphere included methane (CH_4) and ammonia (NH_3).

Modern cells all have a membrane constructed from a phospholipid bilayer. From the very beginning, cells used the lipid bilayer to regulate their internal environment. The bilayer blocked, or impeded, the passive flow of most molecules into the cell, thus protecting the cell from the external environment. Cells exploited this property by embedding proteins in their membranes that would allow only certain molecules to gain entry. In this way, the cell could fine-tune the selection of what got in and what did not. Other proteins embedded in the membrane acted like sensory antennae, making it possible for cells to gain information about their immediate environment. Some of these proteins were used to detect the presence of food molecules, while others became specialized as transmitters and receivers, allowing the cells to communicate with each other. Cell-to-cell communication led to the next stage in the development of life on our planet. Single cells began to form colonies of increasing complexity, eventually transforming themselves into the multicellular creatures that now inhabit the Earth.

The Classification of Cells

The first cells, appearing 3.5 billion years ago, quickly evolved into ancestral prokaryotes and, about 2 billion years ago, gave rise to Archaea, bacteria, and eukaryotes, the three major divisions of life in the world. Eukaryotes, in turn, gave rise to plants, animals, protozoans, and fungi. Each of these groups represents a distinct phylogenetic kingdom. The Archaea and bacteria represent a fifth kingdom, known as the Monera, or prokaryotes. The Archaea are prokaryotes that are physically similar to bacteria (both lack a nucleus and internal organelles), but they have retained a primitive biochemistry and physiology that would have been commonplace 2 billion years ago. Most Archaea are anaerobic and can live in extreme conditions of high temperature (sometimes hot enough to cook an egg) and high salt concentrations. All these conditions were common on Earth 3 billion years ago and make the Archaea seem like living fossils. For the Archaea, oxygen is a toxic substance, and for this reason they are always found living underground, or in deep thermal vents where the concentration of oxygen is very low. There are some bacteria that are also anaerobic but they can tolerate higher concentrations of oxygen than can the Archaea. Eukaryotes



Cell Classification. The first cells evolved into the ancestral procaryotes, which gave rise to the Archaea, bacteria, and eukaryotes, the three major divisions of life in the world. The Archaea and bacteria are very similar anatomically but differ biochemically. Eukaryotes, anatomically and biochemically distinct from both the Archaea and bacteria, gave rise to plants, animals, protozoans, and fungi.

(meaning “true nucleus”) are much more complex than the prokaryotes, having many membrane-bounded organelles and a large genome. These cells are the primary focus of this book and the other volumes in this set.

.2.

PROKARYOTES

Laying the Foundations

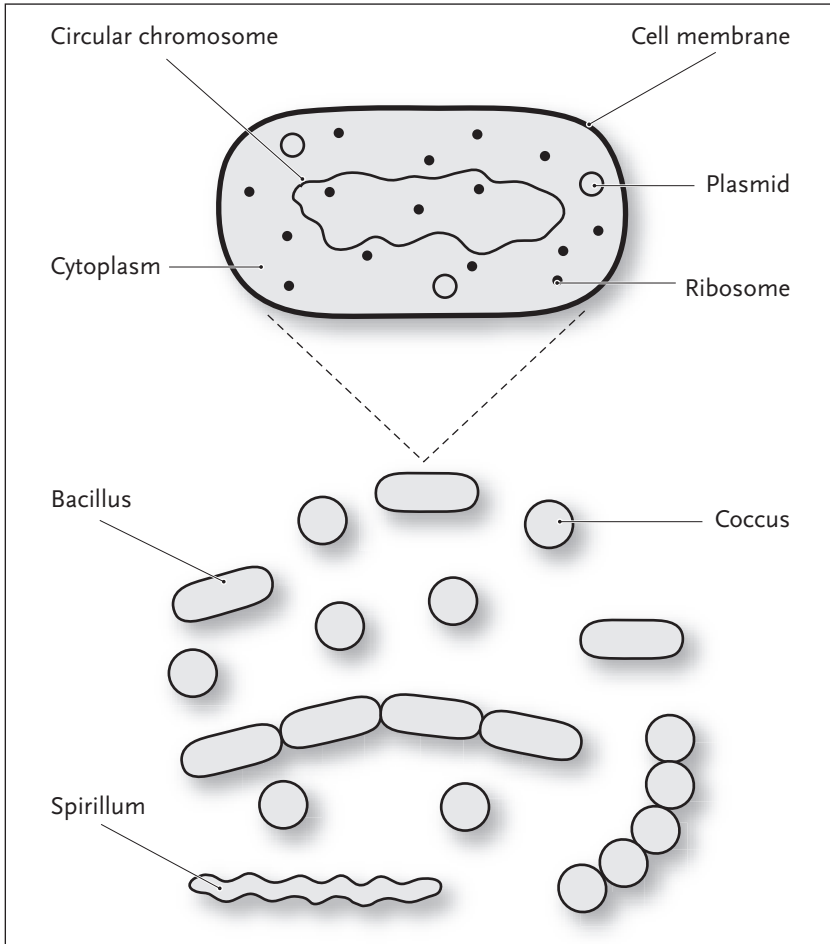
The first cells appeared on earth more than 3 billion years ago and quickly evolved into prokaryotes, more commonly known as bacteria. For more than a billion years, prokaryotes were the only things alive on the face of the Earth. During those billion years, bacteria were likely confined to tide pools and shallow seas, but eventually they came to inhabit virtually every niche available in the water, on the land, and in the air.

A Simple but Versatile Cell

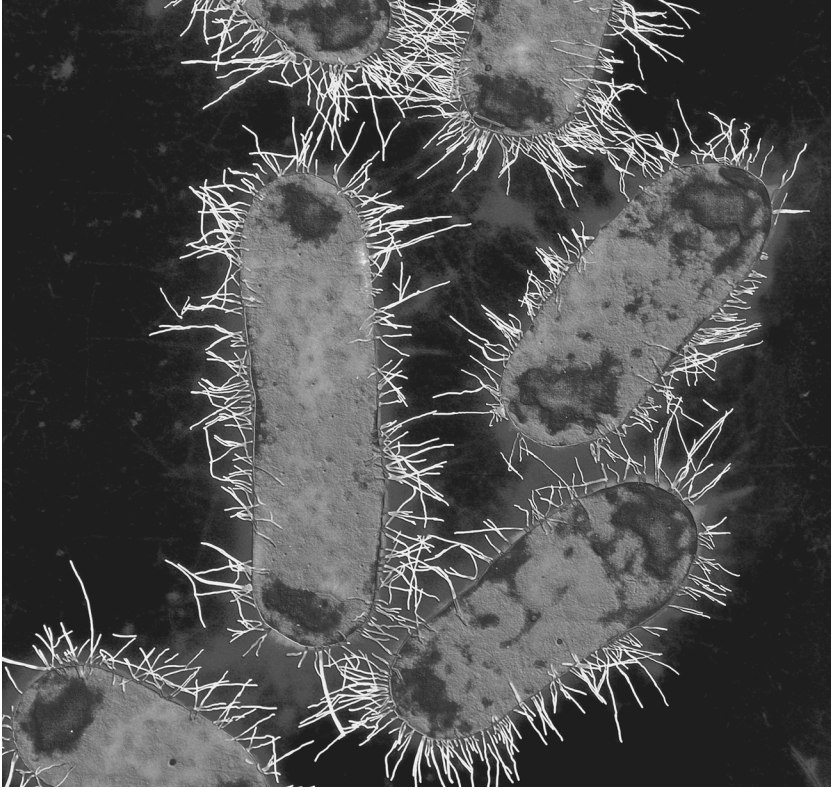
All bacteria are extremely small and invisible to the naked eye. Over a thousand of these cells would fit within a period on this page. No one knew that bacteria, or any other cell, existed until a Dutch lens grinder named Antonie van Leeuwenhoek made the first high-resolution microscope in 1660. Leeuwenhoek's microscope consisted of a single lens mounted in a small brass frame, to which he attached a slender arm for holding a specimen. One September evening in 1683, he looked at a sample of dental plaque taken from his own teeth and the next morning wrote an excited letter to the Royal Society of London describing the many "animalcules" that he had discovered.

Two hundred years later, microscopes and the field of biology had developed to such an extent that more than 1,500 bacterial species had been discovered and described in detail (as described in chapter 8). This wealth of information not only set the stage for the new biology

that was to come but gave us the theoretical framework to understand many diseases that had plagued humankind since the origin of our species.

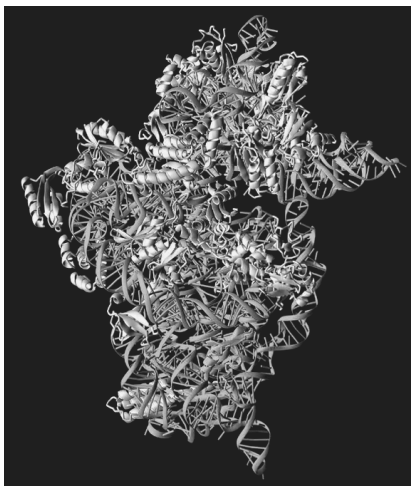


Prokaryotes. All prokaryotes have the same basic anatomy consisting of a cell membrane, cytoplasm and a circular DNA chromosome. Some bacteria have a second, smaller chromosome called a plasmid, which may be present in multiple copies. The cytoplasm contains a wide assortment of enzymes and molecules, as well as ribosomes, protein-RNA complexes that are involved in protein synthesis. The cells may be spherical (coccus), rod shaped (bacillus), or wavy corkscrews (spirillum), appearing singly, in pairs, or linked together into short chains.



Scanning electron micrograph (SEM) of the rod-shaped ciliated bacteria *Escherichia coli*, commonly known as *E. coli*. These bacteria are a normal part of the intestinal flora but certain strains may cause gastroenteritis. *E. coli* is also commonly used in genetic studies. Magnification: 25,000x. (Courtesy of Eye of Science/Photo Researchers, Inc.)

All bacteria have the same simple anatomy, consisting primarily of three parts: a cell membrane, protoplasm (or cytoplasm), and a chromosome. The cell membrane, often surrounded by a cell wall, is a phospholipid bilayer, identical in kind to that which formed around the prebiotic bubbles. The cytoplasm is an aqueous gel that contains a wide assortment of enzymes and molecules, and millions of spherical bodies called ribosomes that are involved in protein synthesis. Prokaryote ribosomes are complex structures consisting of more than 50 different proteins and three RNA molecules. Although the proteins



Molecule model of the 30S ribosomal subunit, which consists of protein (light gray corkscrew structures) and RNA (coiled ladders). The overall shape of the molecule is determined by the RNA, which is also responsible for the catalytic function of the ribosome. (Courtesy of V. Ramakrishnan, MRC Laboratory of Molecular Biology, Cambridge)

outnumber the RNA, two-thirds of the ribosome's mass is due to the RNA. Before ribozymes were discovered, it was assumed the proteins existed in the ribosome as enzymes. We now know, however, that the RNA molecules catalyze the formation of new proteins, while the ribosomal proteins serve a structural role, perhaps acting as a scaffold to hold the amino acids in position before they are linked together.

The bacterial cytoplasm also contains the cell's chromosome: a single circular piece of DNA that holds all the genes, collectively referred to as the genome. Molecular biologists, using recombinant DNA technology (described in chapter 8), have found that a typical prokaryote has 2,000 to 4,000 genes, with each gene coding for a single protein. Many

bacteria have a second, smaller chromosome called a plasmid. Like the main chromosome, the plasmid is circular, but it carries only two or three genes. Plasmid genes have been sequenced and are known to code for proteins that can neutralize antibiotics, such as penicillin or streptomycin. Placing antibiotic-resistant genes on an auxiliary chromosome is a brilliant maneuver. The cell can only have one copy of the main chromosome, but it can have many copies of the plasmid. Consequently, bacteria that have plasmids can produce a large amount of antibiotic-resistant proteins in a very short time. Plasmids make the control of pathogenic bacteria very difficult, but as we will see, their existence was crucial for the development of the new biology.

Because bacteria have such a simple structure, it is often impossible, even under a high-powered microscope, to tell one species from

another. In a few cases, there are clear structural differences: Some bacteria are spherical, whereas others are shaped like short rods. Both types of bacteria can appear singly or linked together into chains of varying lengths. Many rod-shaped bacteria are covered with hairlike cilia or have a single tail-like flagellum to propel the cell through the water. This type of bacterium is said to be motile. Biologists have measured the speed of these cells and have found that if they were the size of a rowboat, they could travel through the water at 60 miles per hour. Spherical bacteria are never ciliated and therefore cannot propel themselves through the water. This type of bacterium is said to be non-motile.

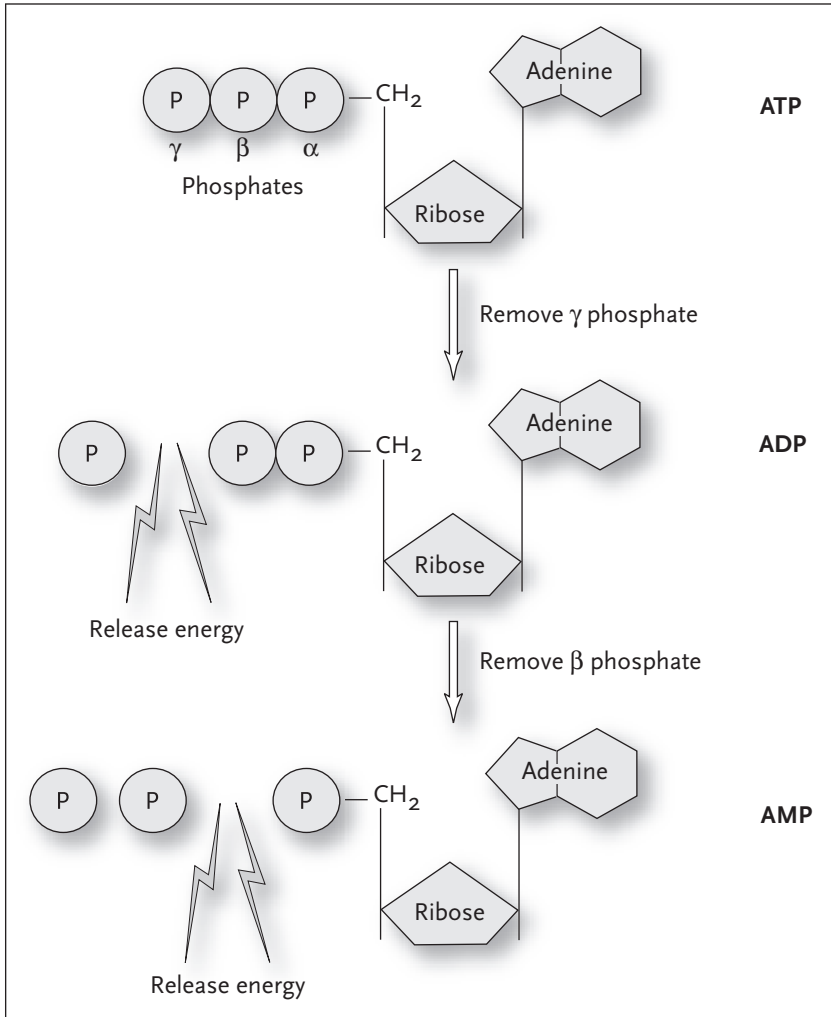
Despite the simple morphology of prokaryotes, their physiology and biochemistry are surprisingly complex. The Earth environment, whether in water, soil, or air, contains an enormous variety of organic molecules, and for each molecule there is a bacterium that is able to use it as a source of food.

The Quest for Power

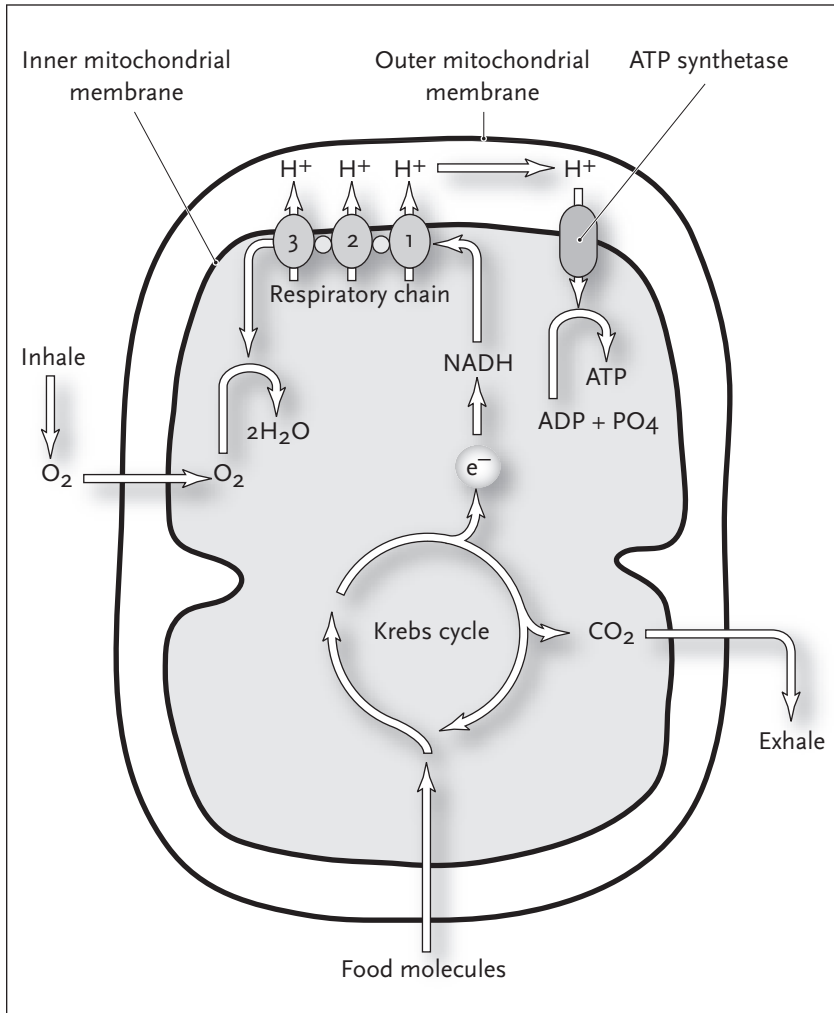
When we strike a match to wood in our fireplace, we gather to enjoy the warmth of the fire, produced by the release of energy stored in the molecular structure of the wood. By striking the match, we are triggering a chain reaction that liberates the energy in one roaring step. In their quest for power, cells learned to avoid roaring steps because they generate so much heat that most of the energy is lost.

Acquiring energy is one of the most important problems that prokaryotes had to solve. The first cells lived in an ocean filled with small molecules, produced in the prebiotic environment. Each of these molecules was like a small piece of wood that the cell could use as a source of energy. The trick was to extract the energy very carefully so as not to lose most of it in a burst of heat and light. Extracting energy in this way is the job of an enzyme. Enzymes can pick a molecule apart piece by piece, supplying small amounts of energy at each stage. The energy so released can be used directly to perform a task, or it can be stored for later use. Extracting energy by breaking a molecule down is called catabolism. Anabolism is just the reverse; it uses energy to build molecules. Cellular metabolism is the combination of catabolic and anabolic activity.

Of all the molecules created in the prebiotic oceans, glucose was the most important as a source of food. Cells prefer glucose to this day, and for some, like the neurons in our brain, it is the only molecule they will



Adenosine triphosphate (ATP), the cell's energy depot. The energy is stored in the covalent bonds linking the phosphates together. Breaking the gamma (γ) phosphate bond releases energy, converting ATP to adenosine diphosphate (ADP). Additional energy may be released by breaking the beta (β) bond, converting ADP to adenosine monophosphate (AMP). AMP is converted back to ATP by mitochondria.



Production of ATP by mitochondria. Food molecules are processed through the Krebs cycle to produce electrons (e^-) that are stored in NADH. The respiratory chain consists of three major components: NADH dehydrogenase (1), cytochrome b (2), and cytochrome oxidase (3). The first component in the chain captures the stored electrons by separating NADH into NAD and H^+ (not shown). The electrons travel through the chain powering a pumping function of each component resulting in a proton (H^+) concentration gradient across the inner membrane and are eventually transferred to oxygen (O_2), leading to the production of water. The protons, moving down their concentration gradient, power the synthesis of ATP by the synthetase. The only exhaust from this power plant is water, which the cell uses, and CO_2 , a gas that is exhaled by the lungs.

use. Prokaryotes can use glucose directly or, in some cases, send it through an anabolic pathway to form a polymer called glycogen (a long chain of glucose molecules), which can be stored in the cell for later use.

Prokaryotes also developed a method for storing some of the energy that is released when glucose is broken down. This procedure is called glycolysis, a very ancient catabolic pathway that produces two molecules of ATP for every molecule of glucose that is broken down. ATP stores energy in phosphate bonds, and this energy is released when these bonds are broken. If an enzyme requires energy for a job that it has to do, it can either break (or hydrolyze) a phosphate bond on ATP, or, if it lacks that ability, the cell recruits another enzyme that has ATPase activity (that is, the enzyme is able to hydrolyze the phosphate bond). Prokaryotes often recruit many enzymes to perform a given task. The complete catabolism of a molecule of glucose by the glycolytic pathway requires 10 different enzymes. Because it does not require oxygen, it is referred to as an anaerobic metabolic pathway.

A half-million years after the first prokaryotes appeared, some of them learned to build organic molecules by using energy collected from the sun. A by-product of this photosynthetic pathway is the release of oxygen. Eventually, other cells, unable to perform photosynthesis, learned to use the oxygen to extract energy from an even wider variety of molecules than was possible with glycolysis. Two aerobic (requiring oxygen) metabolic pathways were developed; one of these is called the citric acid cycle (or the Krebs cycle, after the biochemist who discovered it), and the other is called the electron transport chain (also called the respiratory chain). These two pathways work in tandem to extract energy from fats, simple sugars, polysaccharides, and amino acids.

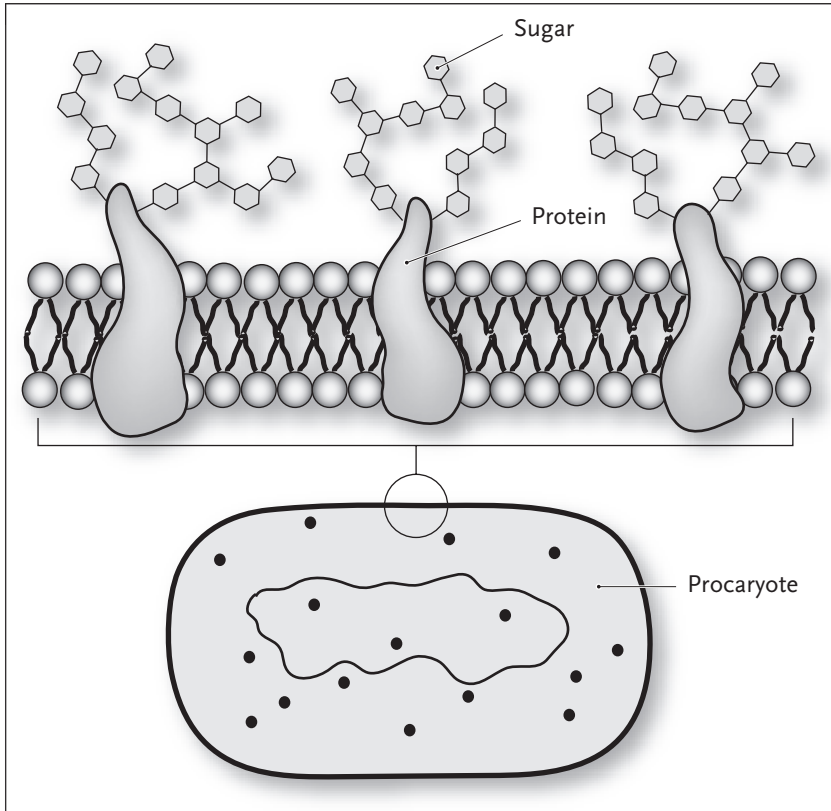
Unlike glycolysis, the Krebs cycle stores most of the energy that it liberates in electrons, which are carried by special molecules through the respiratory chain where their energy is used to make ATP. The by-products of these two pathways are water and carbon dioxide (CO_2). The coordinated activity of the Krebs cycle and the respiratory chain is analogous to the way we generate electricity to run our factories and to make our homes comfortable. A power generator, usually at a hydroelectric dam, plays the role of the citric acid cycle, and the copper wires

that carry the current are analogous to the respiratory chain. We use the electricity our power plants produce to turn on lights, heaters, and motors. The cell uses the electricity that it generates for one thing: to make ATP.

Glycolysis, the Krebs cycle, and the respiratory chain are all run and assembled by protein enzymes. These metabolic pathways are used by all prokaryotes that are alive today. The glycolytic pathway and the Krebs cycle are located in the protoplasm, while the respiratory chain is located in the cell membrane. Additional proteins necessary for collecting glucose and other sugars are also located in the cell membrane. These proteins, called glucose transporters or carriers, are specially designed for bringing glucose into the cell. Glucose and other simple sugars can diffuse passively across the cell membrane, but it is a much slower process. Transporters provide a channel that allows the cell to take up glucose 100 times faster than by simple diffusion.

The ability to make glucose carriers that are embedded in the cell membrane was probably the most important event leading to the transition from the first cells to the ancestral prokaryotes. Once cells learned this trick, they expanded on it very quickly. Proteins were embedded in the membrane that could detect and import other sugars, such as maltose or lactose. They even learned how to make sugar receptors, also embedded in the membrane, which could signal the cell when a high concentration of glucose or maltose was encountered so the activity of the transporters could be stepped up accordingly. The electron transport chain may have evolved from proteins that were originally embedded in the membrane to process or detect sugar molecules.

The sugar carriers, receptors, and components of the respiratory chain are all glycoproteins; that is, sugar molecules are attached to the proteins to enhance or modulate their behavior. Glycoproteins are like molecular trees, with the protein portion being the trunk and the sugar molecules forming the leaves and branches. It is almost as though the prokaryotes were building a forest with which to cover themselves, much in the way higher plants covered the surface of the Earth so many millions of years later. The molecular forest of a prokaryote is called the glycocalyx, and its importance to the cell cannot be overstated. This forest gives the cell its eyes, ears, and a sense of touch in addition to the energy-processing machinery. It is through the glycocalyx that cells



The prokaryote glycocalyx. The glycocalyx is a molecular “forest” consisting of glycoproteins, which covers the outer surface of the cell membrane. In this forest, the “tree trunks” are protein, the “leaves” are sugar molecules, and the “branches” are chemical bonds. Glycoproteins have many jobs, including the transportation and detection of food molecules.

learned how to communicate with one another, paving the way for the appearance of multicellular creatures like ourselves.

The Importance of Good Housekeeping

Acquiring energy is not the only problem that prokaryotes had to deal with. They also had to keep their house in order and be able to make or repair every part of their structure. This was no small feat. By analogy,

if our house is damaged we go to the hardware store to buy the lumber for the repairs; but for a cell there is no such store available. Instead, they have to *make* all the materials needed before they can begin to repair themselves.

Prokaryotes have learned to build and repair themselves with the same molecules that they use for food. As mentioned earlier in this chapter, glucose can be converted into glycogen as a food reserve; after being cross-linked with a few amino acids, it is also used to make the cell wall. Higher plants use a similar trick: Wood is nothing more than an elaborate polymer of glucose. The trees that grow on the Earth and the lumber that is used to build our homes are all sugar compounds.

The prokaryotes use many other molecules both for food and for building materials: Fats can be used as a source of food, or they can be converted to phospholipids and used to repair the cell membrane. Many proteins have enzymatic activity and are kept busy running the cell, but there are also many nonenzymatic proteins, which are synthesized as building materials and used to construct cilia, flagella, microtubules, the cytoskeleton, and all the cellular organelles.

A Lost World

Cells came to life in an RNA world, where RNA functioned both as a molecule for storing genetic information and as an enzyme. But RNA is a relatively small and simple molecule. Like an old-fashioned computer with limited memory, it can store only small amounts of genetic information, and its talents as an enzyme are severely limited. This was no great disadvantage to the first cells, since the molecules they were using as a source of energy were also very simple and did not require enzymes more elaborate than a ribozyme. But that world began to change as the small molecules were used up and replaced with more complex molecules, produced by the cells themselves and liberated into the water when they died. Now the cells were faced with a new problem, a new quest: the extraction of energy and building materials from molecules of increasing complexity.

Cells, always inventive, found a solution. However, there may have been a time when their very survival was in doubt, for it meant they had to change the way they had been doing things for millions of years:

They had to abandon the RNA world in favor of DNA and protein. Protein enzymes, being made from 20 different amino acids, can be much more complex than a ribozyme, and therefore they have a greater chance of being able to catabolize complex molecules. Moreover, as the population of complex food molecules increased, cells were forced to keep pace by coming up with new enzymes to deal with them. RNA could not store the genetic information for all the new protein enzymes that cells needed in order to survive; it is for this reason that DNA came to be the molecule of choice for the storage of genetic information.

A New Genetic Order

In the RNA world, the genetic flow was simple and direct: Ribozymes were both genes and enzymes that could synthesize proteins directly. The prokaryotes, however, changed all that. DNA became the molecule of choice for storing the genes, and RNA was left in control of the protein synthesis machinery. This three-step, unidirectional organization of DNA to RNA to protein is now used by all cells, and has many advantages over the system that it replaced. DNA, being a much longer macromolecule than RNA, can store information for thousands of genes. In addition, because it is a double-stranded molecule, DNA is much more stable than RNA, and the presence of two strands provides a way of repairing the genes if they are ever damaged. That is, because of nucleotide pairing, the damaged strand can be repaired using the complementary strand as a guide.

In the new genetic order, the molecule carrying the gene itself is not being used to synthesize the protein, so there is no chance that it will be damaged during the translation process. There is a real-world analogy to this organization: Human architects and engineers make blueprints to guide them in the production of houses, cars, boats, and many other structures. Those blueprints are always kept in a safe place so nothing will ever happen to them, and the only time they are taken out of the filing cabinet is to make a copy, or a working blueprint, that is given to the carpenters so they can build our new home. The carpenters may spill coffee on it, step on it, or crumple it up if they want to, because they always have the original to refer back to. In the cell, the

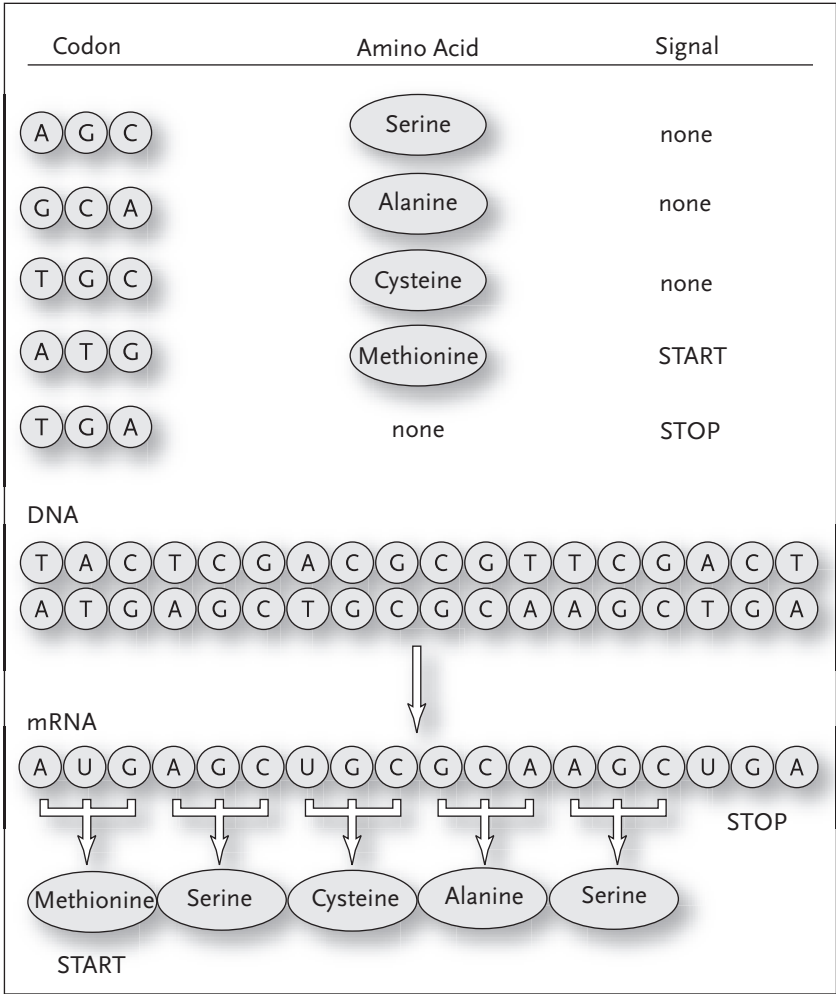
working blueprint is a molecule called messenger RNA (mRNA), so named because it is, in effect, carrying a genetic message from the chromosome to the ribosomes. Copying a gene into mRNA occurs by a process known as transcription. Once the ribosomes receive the mRNA, the ribosomal RNA (rRNA) molecules use it to synthesize the protein in a process known as translation. From this we can see that while RNA lost its job as a genetic reservoir, it remains in control of protein synthesis.

No one knows how DNA came to be the cell's gene bank, but it is possible the enzymatic proteins orchestrated the change themselves. There is a protein enzyme called reverse transcriptase that can copy RNA into DNA, and the reverse transcription process that it performs may have been the way the DNA genome originally appeared. Moreover, protein enzymes are capable of surprisingly complex behavior, almost as though they are living entities unto themselves. In prokaryotes, and in all modern cells, there are proteins that take care of the DNA much like shepherds tending their flocks. Some of these molecular shepherds constantly scan the DNA for damage and, if they find any, repair it. Several protein enzymes are in charge of replicating DNA in preparation for cell division, and still others are needed to copy a gene into messenger RNA in preparation for protein synthesis.

The interaction between the proteins and the DNA is of mutual benefit and important for the survival of the cell. All proteins have a very short life span, sometimes no more than a day or two, so it is to their advantage to take care of the genes. In a sense, this relationship between DNA and protein is a form of molecular symbiosis, in which both parties benefit. The cell as a whole is the ultimate beneficiary, since a good working relationship between the enzymes and the DNA makes it more adaptable and better able to survive in a changing environment.

In building their new genetic order, the prokaryotes had to develop a method for interpreting the sequence information stored in the DNA molecule. DNA is a linear sequence of four different kinds of nucleotides, so the simplest approach would be to have each nucleotide specify a different amino acid. That is, adenine would code for the amino acid glycine, cytosine for the amino acid lysine, and so on. This

simple code may have been useful to the earliest cells, but it is limited to the construction of proteins consisting of only four different kinds of amino acids.



Transcription and the genetic code. Five codons are shown, four specifying amino acids (protein subunits) and two that serve as start and stop signals. The codons, including the start and stop signals, are linked together to form a gene on the bottom, or coding, DNA strand. The coding strand is copied into messenger RNA (mRNA), which is used to synthesize the protein. Nucleotides appear as round beads: Adenine (A), Thymine (T), Cytosine (C), and Guanine (G). Amino acids appear as labeled elliptical beads. Note that in mRNA, uracil (U) replaces the thymine (T) found in DNA.

Prokaryotes undoubtedly experimented with a variety of coding methods before adopting the current system, in which a combination of three out of the four possible DNA nucleotides, called a codon, specifies a single amino acid. With this scheme, it is possible to have a unique code for each of the 20 naturally occurring amino acids. For example, the codon AGC specifies the amino acid serine, whereas TGC specifies the amino acid cysteine. Codons are linked together to form a long continuous sequence that is called a gene. Not all the codons code for amino acids. The sequence TGA signals the end of the gene, and a special codon, ATG, signals the start site, in addition to specifying the amino acid methionine. Thus all proteins begin with this amino acid, although it is sometimes removed once construction of the protein is complete. An average protein may consist of 300 to 400 amino acids; since the codon consists of three nucleotides for each amino acid, a typical gene may be 900 to 1200 nucleotides long. The codon system, developed by the prokaryotes, is also used by the eukaryotes, and for this reason is referred to as the universal genetic code (see table on page 32).

Messenger RNA is the working copy of the gene, but it has the nucleotide uracil in place of thymine, which is found in DNA. Consequently, the DNA codon for methionine is ATG, but it is AUG on the mRNA. The ribosomal RNAs are programmed to recognize the codon as it appears on the mRNA. Once the protein is made, mRNA is broken down and the nucleotides recycled. Thus, mRNA has a very short life span, sometimes referred to as its half-life (the time it takes for half the population to disappear), which can range from a few seconds to a few hours. The short half-life of mRNA makes the cell very responsive to changing conditions in the environment and within the cell itself.

The complete genetic code used by all living things on this planet consists of 64 codons that specify 20 amino acids and the start and stop sites (see table on page 32). The large number of codons is due to redundancy in the code; that is, several codons may specify the same amino acid. Human carpenters, as mentioned previously, use a two-dimensional blueprint to build a three-dimensional house, but prokaryotes have gone us one better. They build three-dimensional objects (proteins), but they do it with a one-dimensional blueprint. Moreover, the invention of the codon, and the use of it to produce identical copies of the same protein over and over again, is nature's first and most important cloning experiment.

THE UNIVERSAL GENETIC CODE		
Codon	Amino Acid	Signal
GCA GCC GCG GCU	Alanine	—
UGC UGU	Cysteine	—
GAC GAU	Aspartic acid	—
GAA GAG	Glutamic acid	—
UUC UUU	Phenylalanine	—
GGA GGC GGG GGU	Glycine	—
CAC CAU	Histidine	—
AUA AUC AUU	Isoleucine	—
AAA AAG	Lysine	—
UUA UUG CUA CUC CUG CUU	Leucine	—
AUG	Methionine	Start
AAC AAU	Asparagine	—
CCA CCC CCG CCU	Proline	—
CAA CAG	Glutamine	—
AGA AGG CGA CGC CGG CGU	Arginine	—
AGC AGU UCA UCC UCG UCU	Serine	—
ACA ACC ACG ACU	Threonine	—
GUA GUC GUG GUU	Valine	—
UGG	Tryptophan	—
UAC UAU	Tyrosine	—
UAA UGA UAG	—	Stop
Codons are written using the standard abbreviation for each nucleotide on the messenger RNA: adenine (A), uracil (U), cytosine (C), and guanine (G). Note that in mRNA uracil replaces the thymine found in DNA.		

The Good, the Bad, and the Ugly

The classification of bacteria is based on the cell’s morphology, its motility, the way they grow on a culture plate, the kinds of sugars and other molecules that they are able to metabolize, and whether their metabolism is anaerobic or aerobic. In 1844, the Danish bacteriologist Hans Christian Gram introduced a simple staining procedure

that provides a convenient way of distinguishing different types of bacteria that otherwise have the same physical appearance. The Gram stain exploits the chemical properties of the cell walls in different species of bacteria. Those bacteria that retain the stain, which is blue, are referred to as being Gram positive and those that do not are Gram negative. For example, *Myxobacteria*, *Staphylococcus*, and *Streptococcus* are Gram positive, whereas *Salmonella*, *Pseudomonas*, and *Cyanobacteria* are Gram negative. DNA sequence analysis of ribosomal RNA has also been used to assign cells to various taxonomic groups.

We generally think of bacteria in the context of a disease they are known to cause, and many of them do cause very serious disorders. But most bacteria fall into the “good” category and can be found living in the water and the soil, on our skin, and in our intestinal tracts. Indeed, there are even bacteria living among the clouds. Because trees and elephants are so big, we tend to think that they, along with other plants and animals, contribute the most to the biomass of the earth, but this is not the case. Bacteria not only outnumber us, they outweigh us by a wide margin.

Bacteria are always busy making a living, just as they have done for billions of years, and the thing that occupies most of their time is the extraction of energy from organic molecules. A by-product of the extraction process is the recycling of nutrients, an activity that is crucial to the maintenance of earth’s ecosystem. Carbon and nitrogen compounds, essential for the synthesis of proteins and nucleic acids, would be lost from the system if not recycled by bacteria. In this regard, the cyanobacteria (sometimes called blue-green algae) are perhaps the most important. This is an extremely ancient group of filamentous bacteria (see table on page 34).

Fossilized remains of cyanobacteria date back nearly 3 billion years and are the oldest fossils known. The discovery of fossilized cyanobacteria was crucial to our understanding of the origin of life and the conditions prevalent on our primitive earth. Cyanobacteria are capable of photosynthesis and are believed to be the evolutionary forerunners of modern-day plant and algal chloroplasts. They are also important members of the phytoplankton in the oceans and freshwater lakes,

CHARACTERISTICS OF SOME COMMON BACTERIA	
Good	
Cyanobacteria	Gram-negative, aerobic, nonmotile short rods that occur singly, in long filaments and irregular colonies. Cells are capable of photosynthesis and live in freshwater and marine habitats. They recycle organic matter.
Myxobacteria	Gram-negative, aerobic, motile rods that occur singly but can aggregate to form fruiting bodies and spores. Cells live in the soil and have a complex life cycle. They recycle terrestrial organic matter and build soil.
Pseudomonas	Gram-negative, aerobic, motile (with several flagella) rods that occur singly or in pairs. Cells live in soil and water and are able to decompose virtually any organic molecule, including crude oil. Important as recyclers and soil builders.
Spirillum	Gram-negative, aerobic, motile (with a single flagella), long spiral-shaped rod that always occurs singly. Cells live in freshwater and marine habitats. They recycle organic matter.
Lactobacillus	Gram-positive, aerobic, non-spore-forming rods that occur singly. Important in the food industry for the production of cheese, yogurt, sour cream, and other dairy products.

accounting for more than 25 percent of the biomass produced in aquatic ecosystems.

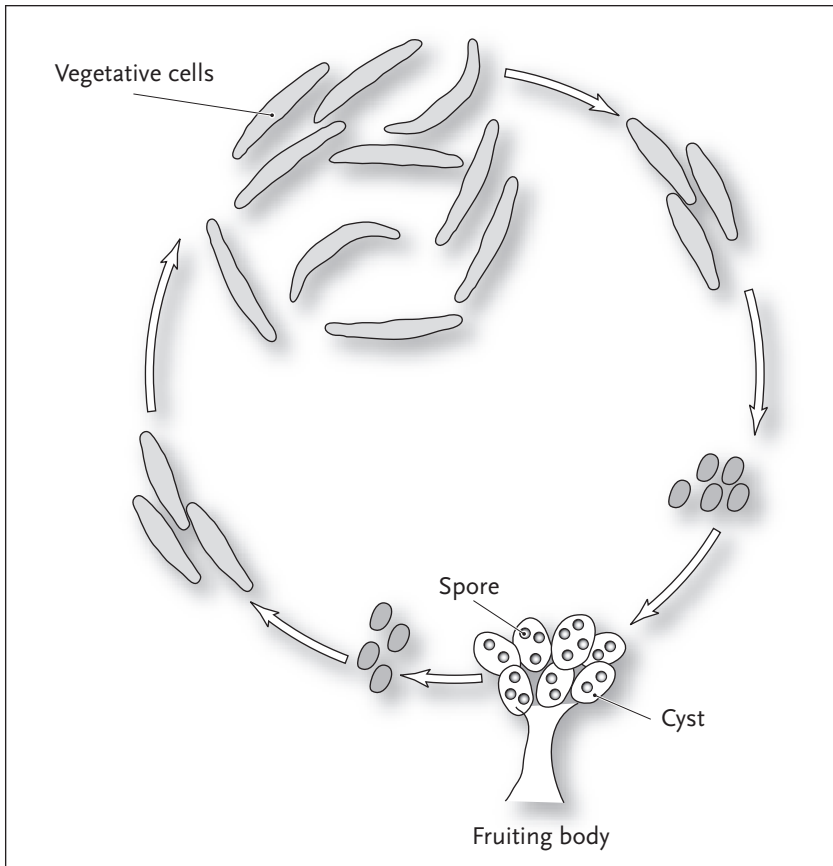
Terrestrial counterparts to the cyanobacteria are the myxobacteria and the pseudomonads, both of which are important for their ability to recycle organic compounds and for making soil. The myxobacteria are especially interesting because of their complex social behavior that is unique among all of the prokaryotes. Myxobacteria live out most of their lives as vegetative cells that browse among the soil and leaf litter. If

Bad	
Salmonella	Gram-negative, aerobic, motile rods that occur singly. Two species, gallinarum and pullorum, cause food poisoning.
Streptococcus	Gram-positive, aerobic cocci (spheres) that occur singly or in short chains. Several species cause throat infections, tonsillitis, and food poisoning.
Staphylococcus	Gram-positive, aerobic cocci that occur singly or in grapelike clusters. <i>S. aureus</i> is a major cause of food poisoning. Symptoms are produced by a toxin.
Ugly	
Bacillus anthracis	Gram-positive, aerobic, nonmotile rods that occur singly. These are spore-forming bacteria that germinate when inhaled by people or cattle. The cells attack the lungs and the disease is usually fatal.
Vibrio cholerae	Gram-negative, aerobic, curved rods that occur singly. Ecology similar to Pseudomonads. Causative agent of cholera, which is usually fatal if not treated promptly.
Neisseria gonorrhoeae	Gram-negative, aerobic, small cocci that occur in pairs. A major cause of sexually transmitted disease, affecting 3 million people a year. Infected pregnant women can pass it to their children, who develop serious eye infections if not treated promptly.

the food supply runs out, the cells aggregate at the top of the leaf litter, where they form a fruiting body that looks like a tiny cluster of grapes. Some of the cells form the stalk and storage capsules, called cysts, while the rest form the spores that become sealed inside the cysts (the grape-like structures). Being on top of the leaf litter exposes the fruiting body to the wind, which may carry some of the cysts to other areas where, hopefully, there is more food available. When the conditions are right, the cysts rupture, releasing the spores, which germinate into vegetative

cells. This remarkable life cycle is an example of the kind of colonial behavior, adopted by primitive cells, that eventually led to the appearance of multicellular creatures.

Not all bacteria live in the soil, air, or water: Many live on the skin or inside the bodies of animals. Those that live inside an animal's body are restricted to the oral cavity, the throat, and the digestive tract. For the most part, these bacteria are harmless, sometimes even beneficial. But there are a few that can make people and animals very sick, and in some



Life cycle of myxobacteria. Vegetative cells feed on organic material in the soil. When food runs low, the cells are stimulated to form a fruiting body consisting of many cysts, which are full of spores. When conditions are favorable, the cysts break open and the spores germinate into vegetative cells.

cases the disease can be fatal. Some of the “bad” bacteria that we have to contend with are species from the *Salmonella*, *Streptococcus*, and *Staphylococcus* genera, which cause food poisoning and, in the case of *Streptococcus*, serious throat infections. Although these bacteria can make people very sick, the diseases they produce are mild compared with those produced by the “ugly” bacteria.

The most notorious among this latter group is *Bacillus anthracis*, a spore-forming bacterium responsible for anthrax, a disease affecting cattle and humans. When an animal inhales *B. anthracis* spores, it is like offering the bacteria a large plate of food. The spores germinate very quickly into vegetative cells and begin consuming the victim’s lung tissue. Death usually follows within a few days. Antibiotics are available for this disease but are usually ineffective unless given to the patient on the day the spores are inhaled.

Bacillus anthracis has recently been used by bioterrorists, who have exploited the deadliness of the disease and the hardiness of the anthrax spores, which can remain viable for years. The spores, which to the naked eye look like a fine powder, were used in 2001 to contaminate letters before they were sent to various destinations in the United States. Several people died of an anthrax infection after being exposed to some of those letters.

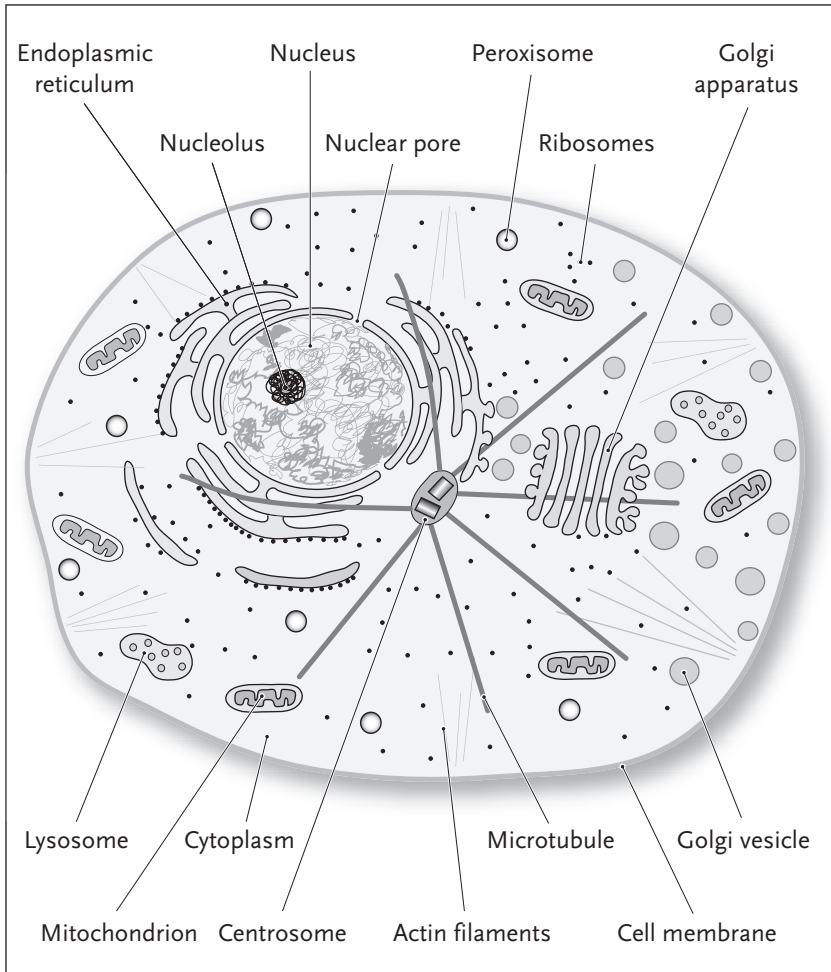
3.

EUKARYOTES

Dawn of a New Era

The only thing in nature more complicated than a eukaryote is a bunch of eukaryotes working together; the heart is one example, the human brain is another. The complexity of these cells came about as a result of necessity and the struggle for survival. Life began in coastal waters that were filled with nutrients, produced by the heat and storms of pre-biotic earth. As the first cells consumed those nutrients, selection pressures led to the appearance of phototrophic prokaryotes, known as autotrophs, that could obtain their energy through photosynthesis. But many prokaryotes, known as heterotrophs, lacked this ability and in order to survive began hunting other prokaryotes for food. Scientists believe the eukaryotes evolved from this latter group of hunting heterotrophs. The new adaptations not only helped the eukaryotes hunt down their prey but also made it possible for those cells to communicate with one another and, eventually, to form colonies and multicellular creatures.

The emergence of the eukaryotes was an important event, not only because a new lifeform had emerged, but also for the survival of the entire ecosystem. Eukaryotes went on to produce enormous populations of photosynthetic organisms both in the water and on the land. Decomposing land plants and animal excretions produced soil, which in turn provides a habitat for billions of prokaryotes and other creatures. These habitats could not have been produced by prokaryotes alone. Thus it is possible that without the eukaryotes, life may have died out on this planet sometime after the first cells consumed the prebiotic nutri-



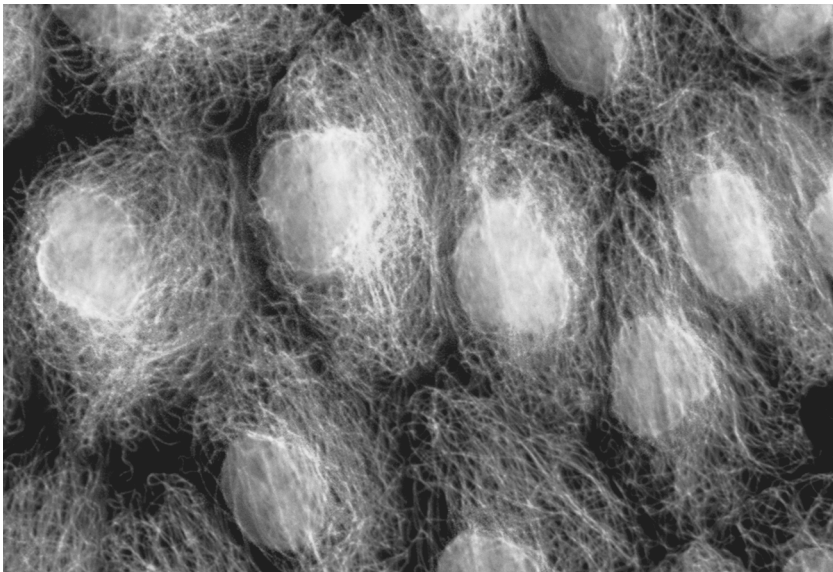
The eukaryote cell. The structural components shown here are present in organisms as diverse as protozoans, plants, and animals. The nucleus contains the DNA genome and an assembly plant for ribosomal subunits (the nucleolus). The endoplasmic reticulum (ER) and the Golgi work together to modify proteins, most of which are destined for the cell membrane. These proteins are sent to the membrane in Golgi vesicles. Mitochondria provide the cell with energy in the form of ATP. Ribosomes, some of which are attached to the ER, synthesize proteins. Lysosomes and peroxisomes recycle cellular material and molecules. The microtubules and centrosome form the spindle apparatus for moving chromosomes to the daughter cells during cell division. Actin filaments, and a weblike structure consisting of intermediate filaments (not shown), form the cytoskeleton.

ents. Eukaryotes took the evolution of life to a new level, and the adaptations that made this possible, and that distinguish them from the prokaryotes, are the subject of this chapter.

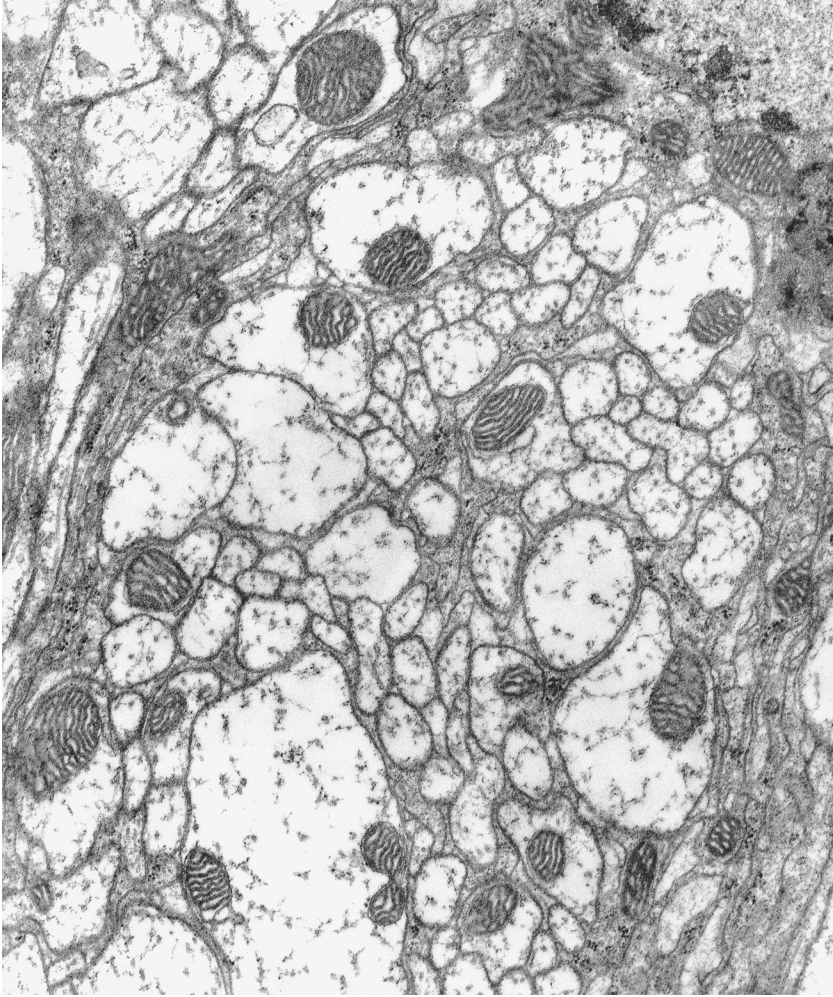
Overview of Eukaryote Structure and Function

Eukaryotes (meaning “true nucleus”) are much more complex than prokaryotes. The prokaryotes keep everything in a single bag. Genes are replicated, transcribed, and translated all in one compartment, the cell’s protoplasm. Eukaryotes changed all this by setting up special membrane-bounded compartments, or organelles, for each job. While the cytoplasm of a prokaryote is plain and homogeneous, the interior of a eukaryote is a maze of organelles, which includes the nucleus, nucleolus, endoplasmic reticulum (ER), Golgi complex, mitochondria, lysosomes, and peroxisomes.

The eukaryote nucleus, bounded by a double phospholipid membrane, contains a DNA (deoxyribonucleic acid) genome on two or more

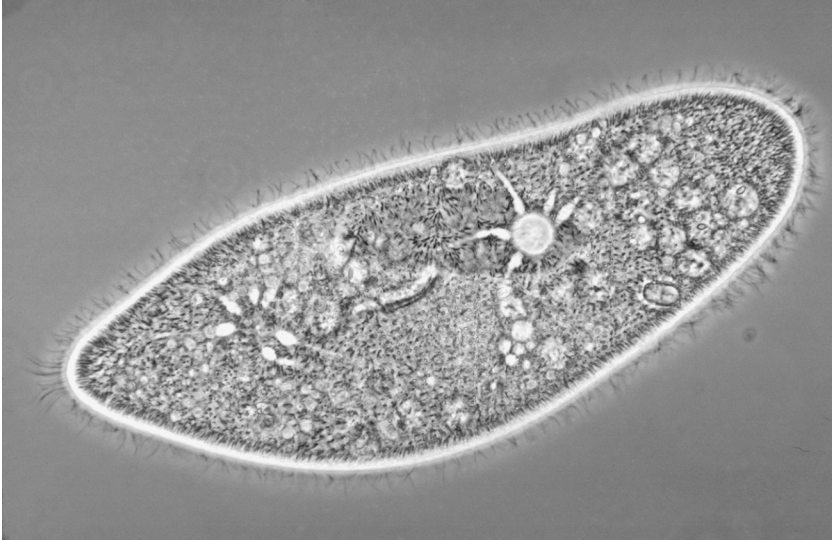


Micrograph of epithelial cells from the pancreas showing the weblike structure of the cytoskeleton. Magnification: 2520x. (Courtesy of Jennifer Waters Shuler/Photo Researchers, Inc.)



Transmission electron micrograph (TEM) of a section of housefly brain showing the large number of mitochondria that is typical of extremely active tissue. Magnification: 19,250x. (*Courtesy of Dr. Joseph Panno*)

linear chromosomes, each of which may contain thousands of genes. The nucleus also contains an assembly plant for ribosomal subunits, called the nucleolus. The ER and the Golgi complex work together to glycosylate proteins and lipids (attach sugar molecules to the proteins and lipids producing glycoproteins and glycolipids), most of which are



Micrograph of the protozoan *Paramecium candatum*. These cells have a characteristic shape of a slipper, and their outer surface is covered in cilia. The cell interior is filled with circular food vacuoles. The bright star-shaped structures are contractile vacuoles, and the macronucleus can be seen lying below the one to the right. Magnification: 160x. (Courtesy of M.I. Walker/Photo Researchers, Inc.)

destined for the cell membrane to form a molecular “forest” known as the glycocalyx. The glycoproteins and glycolipids travel from the ER to the Golgi, and from the Golgi to the cell surface, in membrane-bounded vesicles (bubbles) that form by budding off the organelle by exocytosis. Thus the cytoplasm contains many transport vesicles that originate from the ER and Golgi. The Golgi vesicles bud off the outer chamber, or the one farthest from the ER. Mitochondria, once free-living prokaryotes, and the only other organelle with a double membrane, provide the cell with energy in the form of adenosine triphosphate (ATP). The production of ATP is carried out by an assembly of metal-containing proteins, called the electron transport chain, located in the mitochondrion inner membrane. Ribosomes, some of which are attached to the ER, synthesize proteins. Lysosomes and peroxisomes recycle cellular material and molecules. The microtubules and centrosome form the spindle apparatus for moving chromosomes to the daughter cells during cell division.

Actin filaments, and a weblike structure consisting of intermediate filaments, form the cytoskeleton.

The functional organization of a eukaryote is similar to a carpentry shop. Imagine such a place, where carpenters use a copy of a master blueprint, stored in the shop's office, to build wooden chairs; this shop builds many different kinds of chairs, and it keeps a blueprint for each one. All the machines and tools the carpenters use to build the chairs are located on the shop floor. Every morning, someone in the office takes a blueprint out of the filing cabinet, makes a photocopy of it, and then takes it out to the shop floor and gives it to one of the carpenters. This blueprint may represent a standing order, or the shop may have received an order that morning to produce a special chair. After the chairs are made, some of the carpenters take them into a finishing room, where the frames are painted, after which they go to a shipping room, where they are packaged and sent out to the shop's customers. The energy to power the tools, and to heat and illuminate the shop, comes from electricity, produced at a hydroelectric dam. The shop communicates with its customers, and with other shops, by using the telephone or by sending a letter or an e-mail.

In a eukaryote cell, the shop floor is the cytoplasm and the shop office is the nucleus. The blueprints for everything that the cell makes are kept in the nucleus, but instead of being pictures on a piece of paper, they are made out of DNA. The cell makes proteins of various kinds, and it keeps a blueprint for each one. The blueprints are genes that are arranged, end-to-end, on a very long molecule of DNA. That molecule is the chromosome, and eukaryotes always have more than one.

A eukaryote makes copies of some of its genes every day. Each copy is called a messenger RNA (mRNA), and after being delivered to the cytoplasm, it is used to guide the production of a protein. The cell's carpenters are enzymes that use translation machinery in the cytoplasm to build the proteins. Some of the proteins remain in the cytoplasm to help run the cell, but most are sent to the ER, analogous to the finishing room, where they are glycosylated (painted with sugar), and then to the Golgi complex, where they mature before being packaged for export. These structures, the ER and the Golgi complex, have no counterparts among the prokaryotes. The sugar-coated (glycosylated) proteins from the Golgi are sent to the cell surface, where they form the glycocalyx. As

described in chapter 2, the trunks of the trees in this forest are made from protein and the leaves are made from sugar molecules (the “glyco” part). The glycocalyx is part of the cell’s communication hardware, allowing it to send and receive messages. Cell-surface glycoproteins also form the transporters and ion channels that serve as gateways into the cell. Prokaryotes have a glycocalyx, but its form and function are very simple by comparison. Eukaryotes have refined this hardware to such an extent that a neuron in the human brain can send a signal to the big toe in a fraction of a second. Eukaryotes, like carpentry shops, consume a great deal of energy every day. Carpentry shops get their energy from a hydroelectric dam, whereas eukaryotes get their power from the mitochondria. A hydroelectric dam supplies power in the form of electrons, but mitochondria supply it in the form ATP.

Although organized like a carpentry shop, the internal structure of a eukaryote is more like that of a coral reef. The cell’s “reef” consists of the ER and the Golgi complex; the ER being the larger of the two, fanning out from the nucleus like the rings of Saturn. A real coral reef is a very busy place, with a multitude of colorful fish going about their daily business. Traffic in and around the Golgi complex and the ER is equally hectic, with thousands of multicolored proteins, RNAs, and smaller molecules speeding around as they take care of a thousand chores each day. Traffic between the Golgi complex and the cell membrane alone consists of a constant stream of millions of transport bubbles.

The behavior of eukaryote enzymes, particularly in the nucleus, is akin to that of shepherds tending their flock. Chromosomes, for example, are the most passive macromolecules in the cell, and they are tended by a special group of enzymes that act very much like shepherds. The enzymes move the chromosomes around in preparation for cell division, they regulate the duplication (replication) of each chromosome and the copying of each gene (transcription), and they are constantly inspecting the chromosomes for damage, repairing them when necessary. Why do these molecular shepherds spend so much time and energy caring for the chromosomes? The relationship between DNA and enzymes is an example of the most ancient and most successful cloning experiment. Enzymes have short lives, but a new and exact copy can always be produced as long as the genes are cared for and kept in good shape.

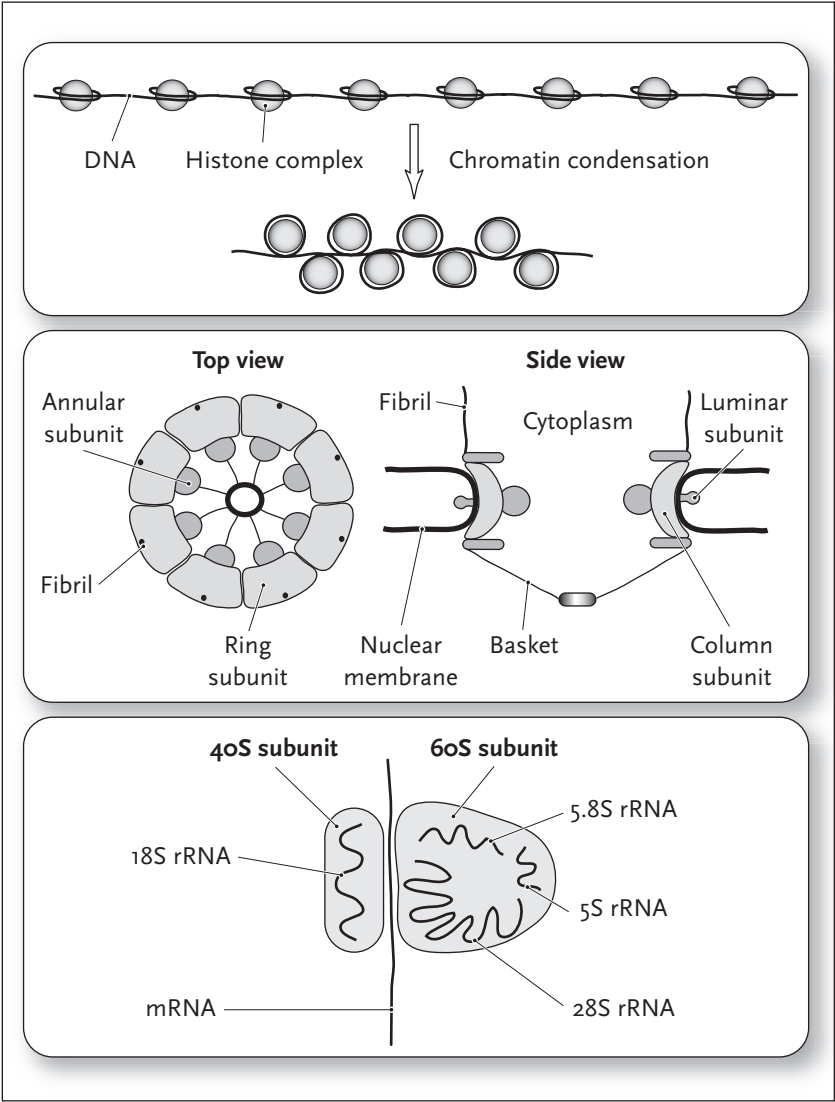
The Nucleus

The nucleus houses the chromosomes and all the enzymes necessary to replicate, transcribe, and repair the genes. The complexity of a eukaryote, compared with a prokaryote, suggests a larger genome, and this has been confirmed with the completion of the human genome project and the sequencing of several prokaryote genomes. Bacteria usually have 2,000 to 4,000 genes, compared with 30,000 in the human genome. This one difference is at the heart of all the other differences between the prokaryotes and the eukaryotes.

Prokaryotes pack all their genes onto a single, ring-shaped chromosome, but a structure such as this could not be used to store 30,000 genes. Eukaryotes solved this problem by splitting their genome into several linear chromosomes. Human cells, for example, have 46 chromosomes, 23 originating from the mother and 23 from the father. Each chromosome is a single, extremely long DNA molecule. This type of chromosome solves the problem of storing a large number of genes, but it becomes a problem when the cell tries to divide. Imagine unwinding and untangling 46 pieces of thread, each of which is several miles long, so they can be laid out, side-by-side, in nice, even rows. The chromosomes have to be aligned in this way for the successful completion of the cell cycle (chapter 4).

Moving long genetic threads around is a major problem but one that eukaryotes solved by coating their chromosomes with special proteins, called histones, that can be manipulated with phosphokinases (enzymes that phosphorylate or add phosphate groups to proteins). Consequently, the chromosome of a eukaryote is not naked DNA, as is that of a prokaryote. There are several kinds of histones, most of which form a spherical structure called a nucleosome that makes the relaxed chromosome look like a string of beads. The complex of DNA and histones is called chromatin. Phosphorylating the nucleosomes is like releasing a stretched rubber band: The chromosome contracts to form a compact structure that is 10,000 times shorter than the bare piece of DNA. Just as a suitcase makes it possible for us to take our clothes on a trip, histones and the chromatin structure they produce make it possible for the cell to package its genes in preparation for cell division.

Chromatin compaction, or condensation, is also used during interphase (the period between cell divisions) to help manage the



Components of a eukaryote cell. Eukaryote chromosomes (top panel) are a complex of DNA and histones, called chromatin, that exists in extended and condensed states. The nuclear pore (middle panel) is an octagonal structure built from many protein subunits that provide a channel through the nuclear membrane. The ribosome (bottom panel) consists of two protein subunits (40S, 60S) that are associated with different ribosomal RNAs (rRNA). Translation is initiated when the two subunits bind to messenger RNA (mRNA).

chromosomes and as one mechanism for controlling gene expression. The packing ratio of interphase chromatin is about 1:1,000 overall, but there are regions where it can be much lower (the density increases as the ratio decreases). This variation in the density of the chromatin accounts for the blotchy appearance of most interphase nuclei. Areas of the nucleus that are very dark present highly compacted chromatin, whereas the lighter regions contain chromatin in a more relaxed state. At the molecular level, chromatin condensation is an extremely dynamic process that is used to close down single genes or whole neighborhoods consisting of hundreds of genes. The mechanism by which this occurs is fairly straightforward: highly condensed chromatin blocks the transcription machinery, so it cannot get access to the gene. Details of this mechanism, and others that control gene expression, will be covered in chapter 5.

Keeping the genes in one compartment and the protein manufacturing machinery in another compartment requires a constant flow of traffic across the nuclear envelope. This traffic is not just the mRNA moving out to the cytoplasm; it also includes the millions of proteins that are synthesized in the cytoplasm but stationed inside the nucleus. Histones, for example, must be imported at a rate of 1 million every three minutes. DNA replication and repair enzymes, and the protein components of the ribosome, which are assembled in the nucleolus, are also part of the traffic.

Eukaryotes evolved an adjustable nuclear pore with an exquisite octagonal geometry to cope with all the traffic moving in and out of the nucleus (see the figure on page 46). The pore is constructed from four subunits, each of which is composed of more than one kind of protein, and basketlike assembly on the nuclear side. All together, nearly 50 different kinds of protein are used to construct this pore, and a typical cell has 3,000 to 4,000 of them, evenly spaced over the surface of the nuclear envelope. At certain magnifications, the pores make the nuclear envelope look like the crater-marked surface of the moon (although, unlike moon craters, nuclear “craters” are all the same size).

The nuclear pore blocks the passage by simple diffusion of any molecule larger than nine nanometers (nm, 10^{-9} meters) in diameter. Nuclear proteins that are larger than this gain access by a specific interaction with the cytosolic fibrils that causes the pore to open, somewhat

like the iris diaphragm of a camera, just enough to let the molecule in. Scientists know the interaction between the protein and the fibril involves a nuclear localization sequence (NLS) on the protein itself, and without it, the pore will not allow the protein to pass. The NLS is like a password or a passport written in the language of an amino acid sequence, usually located at the very end of the protein. A common password for a nuclear protein is Lysine-lysine-lysine-arginine-lysine. If a nuclear protein presented itself to a nuclear pore with a mutated password, such as Lysine-threonine-lysine-arginine-lysine, it would not get in. Scientists have been able to trick the nuclear pore by adding the correct password to a protein that does not normally belong in the nucleus. Such experiments have made it clear that the pore checks the password only and does not care about the identity of the protein. The mechanism by which the pore is able to recognize the correct password and vary the diameter of its channel accordingly is still a mystery.

Protein Synthesis

Eukaryotes synthesize their proteins using ribosomes and the genetic code, just as prokaryotes do. But (as explained in chapter 2) ribosomes are constructed from a mixture of protein and RNA, and this fact created a problem for the eukaryotes. The RNA is made in the nucleus, but the protein is made in the cytoplasm: Where should the ribosome be made? The RNA could be synthesized and shipped out to the cytoplasm, where the ribosome could be assembled, or the protein components could be shipped into the nucleus for final assembly. But given that mRNA is also synthesized there, assembly of the ribosome in the nucleus could lead to a real disaster. If the ribosomes encountered the mRNA, the cell would end up synthesizing proteins in both the nucleus and the cytoplasm, which is exactly what the two compartments were designed to prevent. What a quandary! One solution would be to remake the ribosome so it is all protein or all RNA. However, this would be a formidable task because the ribosome is a very complex structure that took many millions of years to evolve.

To appreciate the complexity of the ribosome, let us examine its components. The eukaryote ribosome consists of two major complexes, called the 40S and 60S subunits (the S refers to their rate of sedimentation

in a centrifuge; the larger the S value, the larger the rRNA). The 40S subunit consists of 33 proteins and a single 18S (having 1,900 nucleotides) ribosomal RNA molecule (rRNA). The 60S subunit has 50 proteins and three different rRNA molecules that range in size from 120 nucleotides (5S) to 4,700 nucleotides (28S). A functional ribosome is produced when the 40S and 60S subunits bind to an mRNA. After translating the mRNA to produce a protein molecule, the ribosomal subunits dissociate from the mRNA (see figure on page 46).

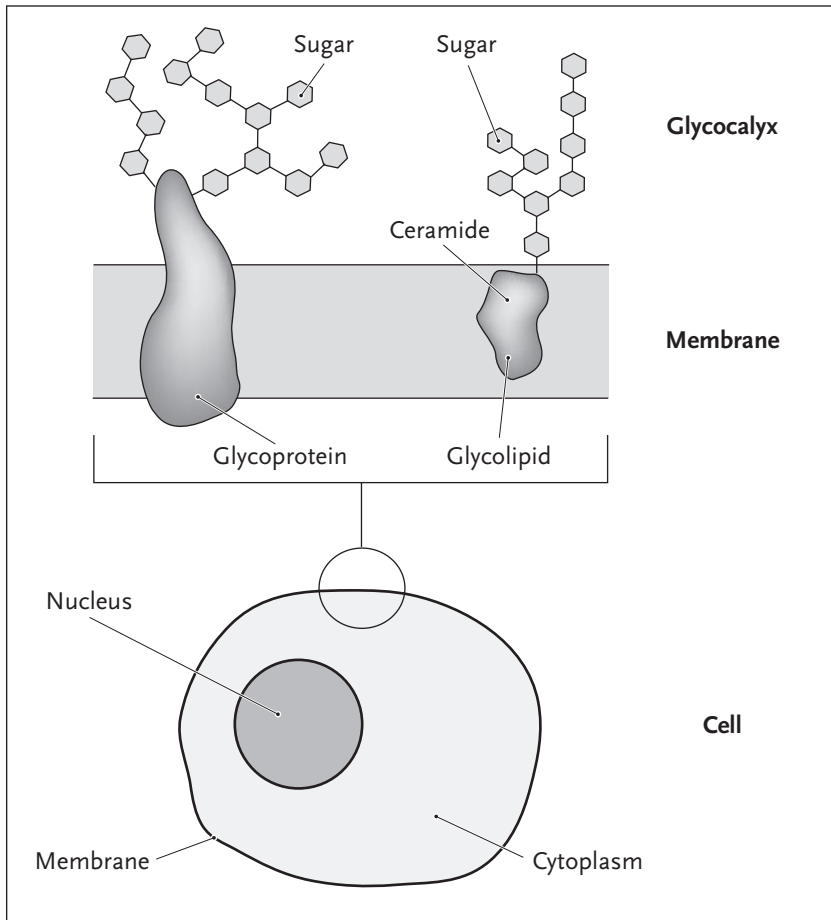
Trying to re-create such a complex structure is out of the question. Instead, the eukaryotes settled on a compromise strategy to keep the ribosome as it is (although they did add a few extra proteins and one extra RNA molecule), and to assemble the subunits in the nucleus but not join them together there. Subunit assembly occurs in the nucleolus, where the rRNAs are synthesized and where ribosomal proteins collect after entering the nucleus from the cytoplasm. Once assembled, the inactive subunits can be safely transported to a pore. Biologists believe the final activation of each subunit occurs as they pass through separate nuclear pores. Once they are in the cytoplasm, they are free to associate with mRNA to initiate translation.

Designing a Molecular Forest

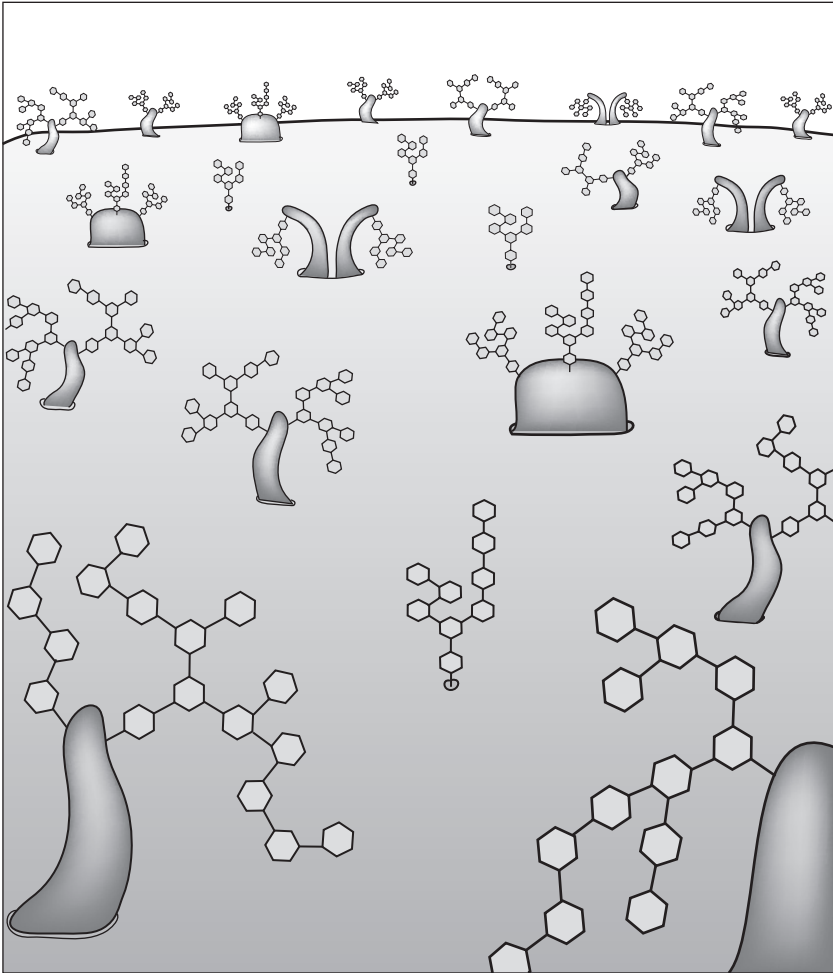
The eukaryote glycocalyx is more complex than the one produced by prokaryotes. Prokaryotes use the glycocalyx to collect and import food molecules, whereas the eukaryotes use it both for food collection and for cell-to-cell communication. Consequently, the eukaryote glycocalyx has a much more diverse collection of glycoproteins embedded in the membrane, and there is a higher proportion of glycolipids, particularly in nerve cells. The production of glycoproteins in prokaryotes occurs in the cytoplasm, but in eukaryotes they are produced in the ER and the Golgi complex.

Proteins destined for the cell membrane carry a marker sequence, much in the way nuclear proteins carry an NLS. When the ribosome detects the marker sequence, it moves to the surface of the ER where it threads the protein through a pore as it is being synthesized. Ribosomes, located on the surface of the ER, can be clearly seen in electron micrographs; those areas are referred to as the rough ER. Once the protein is

inside the ER, it is glycosylated by several different enzymes that add the sugar molecules sequentially. This is analogous to a team of painters working on the same canvas. One painter might lay in the sky and



The eukaryote glycocalyx. The eukaryote's molecular forest consists of glycoproteins and glycolipids. Two examples are shown at the top, a glycoprotein on the left and a glycolipid on the right. The glycoprotein trees have "trunks" made of protein and "leaves" made of sugar molecules. Glycolipids also have "leaves" made of sugar molecules, but the "trunks" are a fatty compound called ceramide that is completely submerged within the plane of the membrane. The glycocalyx has many jobs, including cell-to-cell communication, and the transport and detection of food molecules. It also provides recognition markers so the immune system can detect foreign cells.



A panoramic view of the glycocalyx. The glycoproteins in the cell's forest come in many different shapes and sizes, and they dominate the surface of most cells. The glycolipids all have the same ceramide trunks, but the molecular foliage varies considerably. All but four of the structures in this image are glycoproteins, but in nerve cells they are much more common.

ground, after which another paints the clouds and rocks. The glycosylating enzymes seem to follow a set of rules, because the same kinds of glycoproteins are produced over and over again, but the nature of those rules is still unclear. When the enzymes in the ER are finished, the glycoprotein is loaded into a transport vesicle (bubble) and sent to the Golgi complex.

The Golgi complex has its own group of enzymes that specialize in refining the process begun in the ER. They begin by adding more sugar molecules, and as the glycoprotein (now looking very much like a molecular tree) is modified by successive enzymes, it moves from one Golgi chamber to the next, each time being transported by a Golgi vesicle. In each chamber, enzymes add, trim, or prune the molecular leaves on the trees. Again, as with ER glycosylation, the decision-making process guiding these tasks is poorly understood. When the enzymes are satisfied with the shape and appearance of the glycoprotein, it is loaded into a Golgi vesicle and sent out from the *trans* Golgi network (the outermost chamber) to the cell membrane. Fusion of the vesicle membrane with the cell membrane automatically plants the molecular tree.

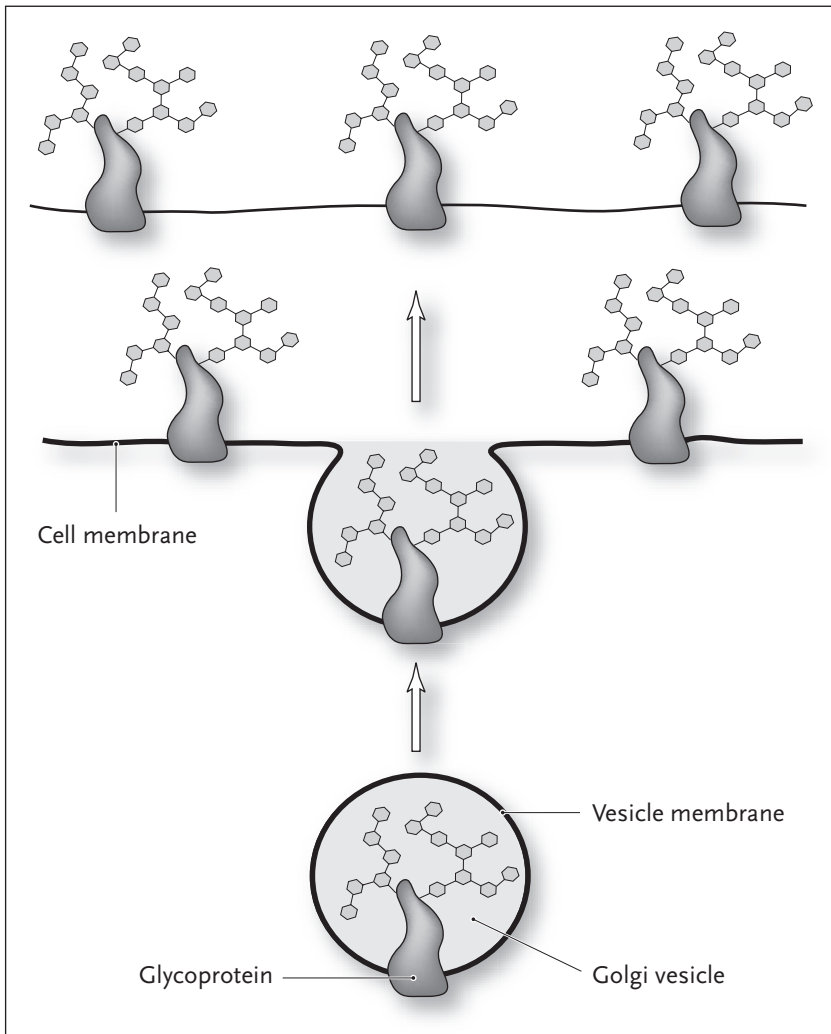
Not all the glycoproteins that pass through the ER or Golgi are destined for the cell membrane. Many are intended for the lysosomes, and some are secreted. Indeed, Golgi vesicles carrying lysosomal glycoproteins become that organelle, as will be described later in this chapter.

The Birth of Communication

Eukaryotes expend a tremendous amount of time and energy building and maintaining the glycocalyx, a molecular forest that would, if we were small enough to walk through it, rival the beauty of any forest on the surface of the Earth. Despite this, eukaryotes build the glycocalyx not because they want to look good but because they need to communicate with their environment, and especially with other cells. The glycoproteins in the cell's forest come in many different shapes and sizes, but functionally they all fall within one of three groups: transporters, ion channels, and receptors.

Transporters are designed to carry specific molecules (usually food of some kind) across the cell membrane. Each type of molecule, such as glucose, lactose, and amino acids, has its own transporter. The intestinal tract, which absorbs the nutrients from the food we eat, has an enormous population of each kind of transporter. Virtually all the nutrients are taken into intestinal cells by transporters and then released into the blood by transporters working in reverse, that is, moving molecules

from inside the cell to the outside (into the circulatory system). Other tissues of the body also have transporters. Most animal cells use glucose as their primary source of energy; this is particularly true of the brain. Consequently, all the tissues in an animal's body have a great number



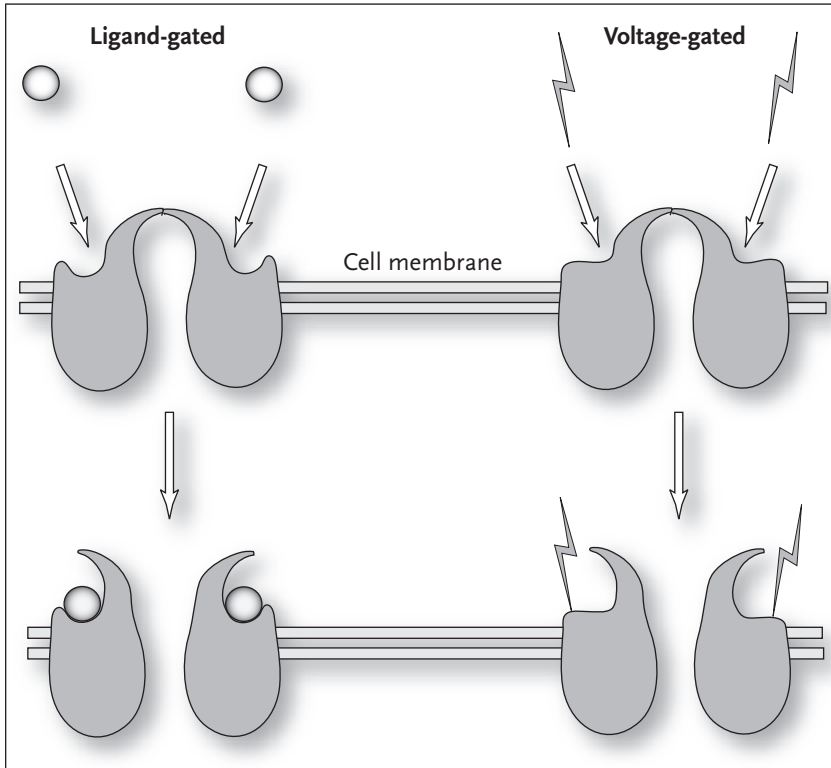
Planting a molecular forest. Vesicles from the Golgi complex carry glycoproteins to the cell surface. Fusion of the vesicle membrane with the cell membrane automatically plants the molecular tree.

of glucose transporters (other sugars, brought in by the intestinal cells, are converted to glucose before being released into general circulation, so other kinds of sugar transporters are not needed outside the intestinal tract).

The second group of glycoproteins, the ion channels, represent a very powerful messaging system that uses the flow of ions into the cell as a means of communication; much in the way we use the flow of electrons down a phone line to speak with someone at a great distance. Ion channels are gated, that is, they can be opened and closed. Ligand-gated ion channels open when a signaling molecule binds to the channel. Voltage-gated channels open when they sense, or are jolted by, an electric field across the membrane.

Receptors, the third group of cell-surface glycoproteins, are the most diverse group of cell-surface glycoproteins and, along with ion channels, are at the heart of the cell's ability to communicate with other cells. Each receptor is designed to respond to a specific signaling molecule. When a signaling molecule binds to its receptor, it sets off a chain of biochemical events in the recipient, or target cell. The signal may cause the cell to increase or decrease the output of a product that is required by other cells in the body, or it may order the cell to grow, to stop growing, or even to commit suicide.

The most common type of receptor that cells use for communication is the G-protein linked receptor. A G-protein is an enzyme that gets its power from an energy reservoir called guanosine triphosphate (GTP), a close relative of the cell's main energy reservoir, adenosine triphosphate (ATP). Binding of a signaling molecule to one of these receptors leads to the activation of the G-protein, which in turn activates a transduction, or signal conversion, enzyme. This enzyme, in turn, produces or activates secondary messengers that cause the cell to carry out the appropriate response. This chain of events, from the moment the signaling molecule binds to its receptor, is called a signal transduction pathway. Alternatively, some signaling pathways use an enzyme-linked receptor, effectively cutting out the G-protein middleman. In a pathway such as this, the receptor itself is also a transduction enzyme, which is activated by the binding of a signaling molecule. The activated receptor phosphorylates, and thereby activates, secondary

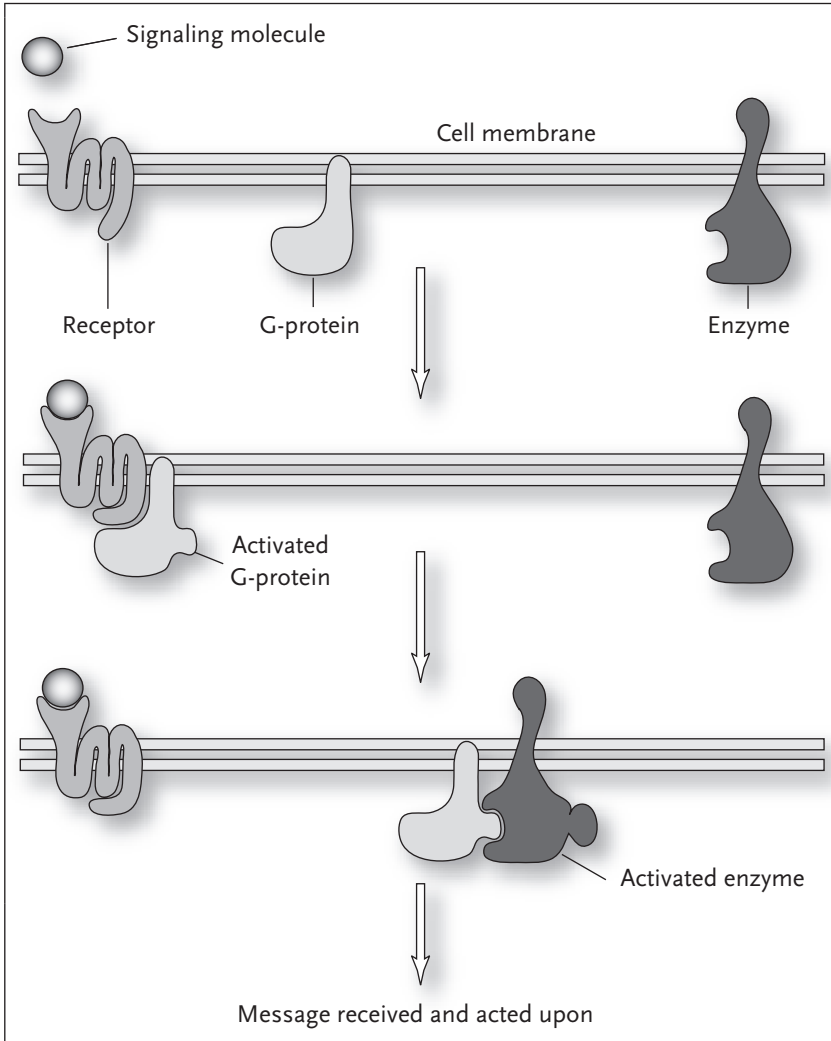


Ion channels. Ligand-gated channels open when bound to a signaling molecule, and voltage-gated channels open when they receive an electric jolt. Ion channels are glycoproteins that are an important part of the glycocalyx.

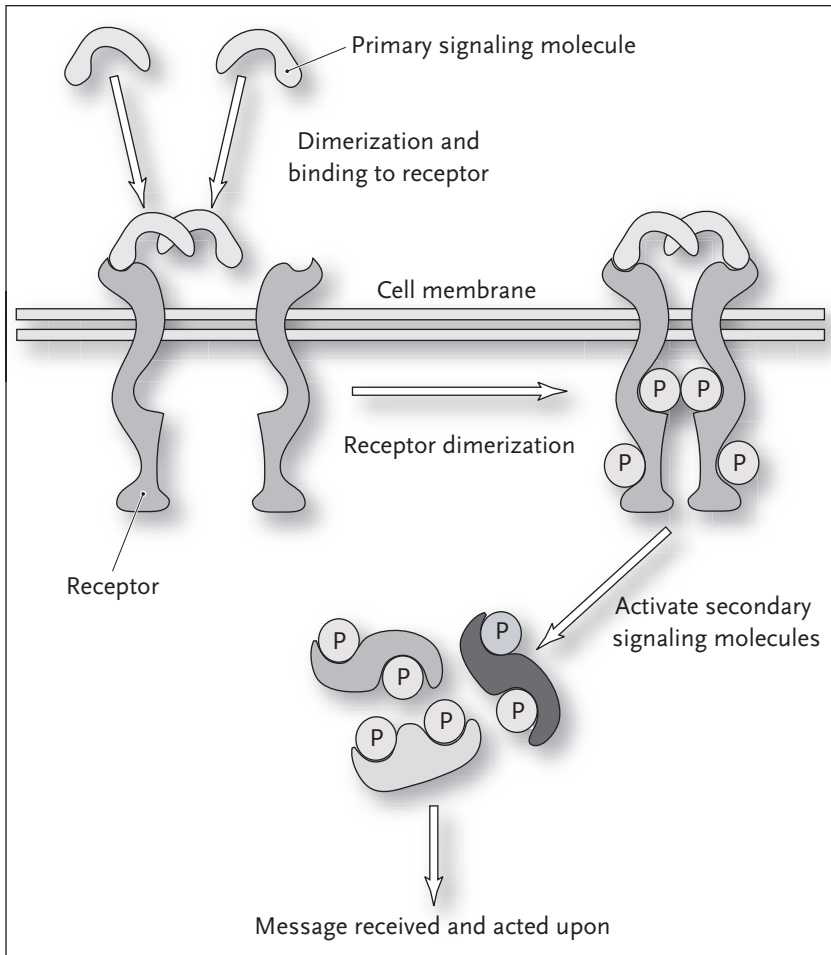
messengers that lead to a change in the activity or behavior of the target cell.

In a multicellular organism, such as a mouse or a human, communication is taken care of by the nervous system, using a combination of ligand-gated and voltage-gated ion channels, or by the endocrine system, which releases hormones as signals to other cells. These two systems coordinate virtually everything that goes on in an animal's body. The exquisite regulation of ion channels in neurons gives us our intellect, vision, hearing, and all other sensory perceptions. The endocrine system regulates our growth, sexual maturation, general energy levels, and even

our mood from one day to the next (the function of neurons, and cell communication in general, will be described in greater detail in chapter 7). The regulation and coordination of ion channels for the purpose of cell-to-cell communication has no counterpart among the prokaryotes.



G-protein-linked receptor. The signaling molecule binds to the receptor, leading to the activation of the G-protein, which in turn activates an enzyme. The enzyme activates secondary messengers, which effect some change in the cell.



Enzyme-linked receptor. The primary signaling molecule forms a dimer (two identical molecules bound together) and binds to the receptor, stimulating dimerization of the receptor. This stimulates phosphokinase activity of the receptor, which phosphorylates itself and several secondary signaling molecules. The activated signaling molecules effect some change in the cell.

Symbiosis and the Quest for Power

As described in chapter 2, prokaryotes developed two aerobic systems for extracting energy from food molecules: the Krebs cycle, which stores most of the energy as electrons, and the electron transport chain, which

uses the energy to make ATP. In eukaryotes, both processes occur in an organelle called the mitochondrion. Consequently, these organelles are responsible for providing the cell with the ATP it needs to power all its biochemical reactions (although a small amount of ATP is provided by glycolysis, which is carried out in the cell's cytoplasm).

For many years, cell biologists believed the mitochondria originated with the eukaryotes, but in 1910, the Russian biologist Konstantin Mereschkovsky suggested that these organelles were once free-living prokaryotes. The earliest evidence in support of this notion, presented by the American biologist Lynn Margulis in 1970, focused on the fact that mitochondria have their own genome and are morphologically similar to bacteria. Since then, Margulis and others have described several interesting similarities between mitochondria and prokaryotes:

- Mitochondria divide by simple binary fission, as do bacteria.
- Synthesis of ATP occurs through the same pathway used by bacteria.
- Mitochondria and bacteria both have a single, circular DNA chromosome.
- The mitochondrial genome is very similar in structure and organization to that of many bacteria.

Today most scientists accept the symbiogenesis theory, which states that the evolution of eukaryotes was associated with the acquisition of symbionts, particularly mitochondria (in animals) and chloroplasts (in plants), about 2 billion years ago. Consequently, mitochondria have been handed down from one generation to the next ever since that time. In addition, for all the animals in the world, inheritance of mitochondria is through the maternal line. We get our mitochondria from our mothers. This fact has been applied by scientists at the University of California, Berkeley, who traced the origins of the human species to a single “mitochondrial Eve” who apparently lived in Africa 100,000 to 200,000 years ago.

Recycling and Defense

The lysosome is a cell organelle that is responsible for recycling worn-out cell parts, such as mitochondria, membranes, and various molecules. Lysosomes are vesicles that are filled with powerful enzymes that

can break down virtually any structure or component of the cell. Lysosomal enzymes are glycoproteins that are sent through the ER and Golgi complex, but the Golgi vesicles containing these enzymes stay in the cytoplasm (rather than fusing with the cell membrane), where they mature into functional lysosomes.

Lysosomes are also very important for maintaining the cell membrane and other membrane-bounded structures. Traffic through the ER and the Golgi complex results in the fusion of ER vesicles with the Golgi and the fusion of Golgi vesicles with the cell membrane. If steps were not taken to recycle parts of both membrane systems they would grow very large, very quickly. In the case of the cell membrane, regular rounds of endocytosis serve to keep the membrane at a constant size. The endocytotic vesicle is fused with a lysosome where the parts are broken down, with some being returned to the Golgi and ER.

Cellular defense among animals is usually left to the immune system. But each cell does have built-in defense capabilities, left over from the time when cells were free living. This defense mechanism involves a coordination between the lysosomes and endocytosis. One form of endocytosis, called phagocytosis, makes it possible for the cell to engulf an invading bacterium. The phagocytic vesicle containing the bacterium fuses with a lysosome and breaks down the microbe to individual molecules, which are released into the cytoplasm or expelled from the cell. All eukaryotes have this defensive capability, but it is used most effectively by macrophages, an important cellular member of an animal's immune system.

The emergence of the eukaryotes came with the appearance of a super cell, about 2 billion years ago, that was 10 times larger than an average prokaryote. In addition, eukaryotes have 10 times the number of genes found in a typical prokaryote, and a complex and highly organized set of organelles that maximize the efficiency of basic manufacturing processes. The main priority of a eukaryote is the construction of a highly complex glycocalyx. The eukaryote glycocalyx greatly improves the efficiency of food gathering, particularly for single-cell eukaryotes that have a predator lifestyle. The improved glycocalyx also made it possible for eukaryotes to communicate with each other, and this capability paved the way for cell colonies and, eventually, true multicellular organisms.

.4.

THE CELL CYCLE

Cells inherited the power of reproduction from the prebiotic bubbles that split in half at regular intervals, under the influence of their turbulent environment. This pattern of turbulent fragmentation followed by a brief period of calm is now stuck in the molecular psyche, or behavior pattern, of every cell. Even today, after 3 billion years, many bacteria still divide every 20 minutes. Though eukaryotes take longer to divide, their regularity is just as profound.

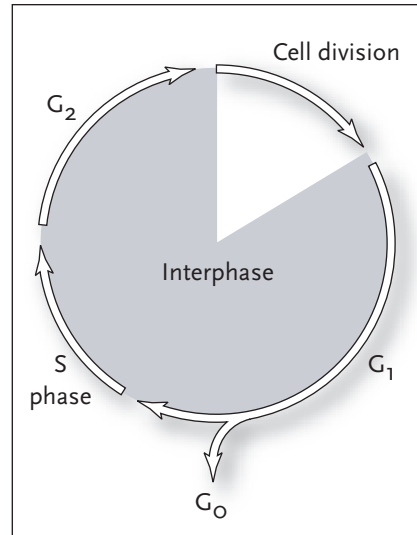
The regular alternation between division and calm has come to be known as the cell cycle. In studying this cycle, scientists have recognized different states of calm and different ways in which a cell can divide. The calm state of the cell cycle, referred to as interphase, is divided into three subphases called Gap 1 (G_1), S phase (a period of DNA synthesis), and Gap 2 (G_2). The conclusion of interphase, and with it the termination of G_2 , occurs with the division of the cell and a return to G_1 . Cells may leave the cycle by entering a special phase called G_0 . Some cells, such as post-mitotic neurons in an animal's brain, remain in G_0 for the life of the organism.

Although interphase is a period of relative calm, the cell grows continuously during this period, working hard to prepare for the next round of division. Two notable events are the duplication of the spindle (the centrosome and associated microtubules), a structure that is crucial for the movement of the chromosomes during cell division, and the appearance of an enzyme called maturation-promoting factor (MPF) at the end of G_2 . MPF is the enzyme that phosphorylates the histones in order to compact the chromosomes in preparation for cell division. MPF is also responsible for the

breakdown of the nuclear membrane. When cell division is complete, MPF disappears, allowing the chromosomes to decondense and the nuclear envelope to re-form.

Many of the events that occur during interphase are poorly understood, but we do know the intention of all the labor is to ensure the production of two identical daughter cells. When the prebiotic bubbles divided, the two new bubbles were not necessarily identical to each other, as there were no mechanisms in place to ensure an equal distribution of the parent bubble's contents. The dissimilarity of the daughter bubbles was an advantage at that stage of development, but true cells had too much to lose to permit haphazard divisions. Thus, the appearance of true cells required mechanisms that guaranteed identical daughter cells.

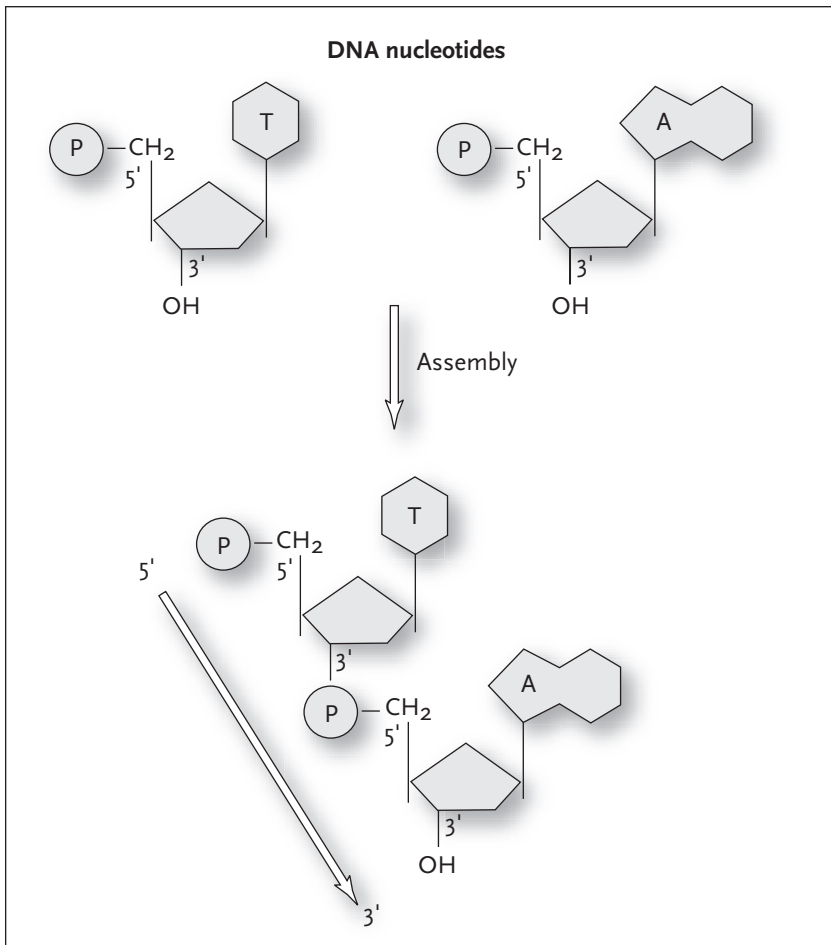
Natural selection is the driving force behind the appearance of novel cellular mechanisms. If a cell happened on to a useful biochemical reaction, there would have been a strong selection pressure to preserve that innovation by providing a way for that cell to hand it down to succeeding generations. The acquisition of new traits and abilities is always linked to the appearance of enzymes with new talents. The earliest cells had already begun storing genetic information for their enzymes, first as an RNA genome and later as DNA. The problem they faced initially was the need to duplicate their genome accurately and to coordinate that process with cell division in order to ensure the production of identical daughter cells.



The cell cycle. Most cells spend their time cycling between a state of calm (interphase) and cell division. Interphase is further divided into three subphases: Gap 1 (G_1), S phase (DNA synthesis), and Gap 2 (G_2). Cells may exit the cycle by entering a special phase called G_0 .

DNA Replication

To understand how a cell duplicates its genome during S phase, it is necessary to understand something about the structure of a nucleotide and the way in which several nucleotides are linked together to form a single-stranded DNA molecule. Nucleotides, shown in the figure below, are linked together to form single-stranded

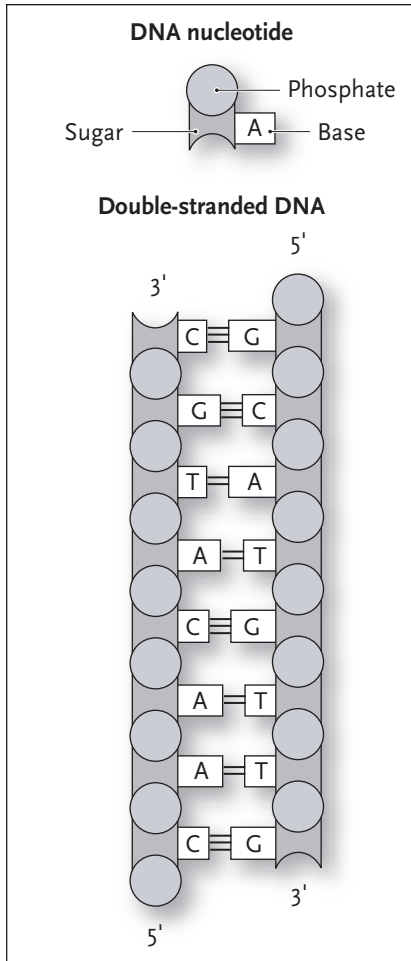


Nucleotide linkage. Nucleotides are linked together in a 5' to 3' direction by linking the phosphate of the adenine (A) nucleotide to the 3' carbon of the thymine (T) nucleotide. The OH group on the T nucleotide is lost in the process.



Computer model of DNA. The two strands coil around each other to form a helix that, when looking down on it from above, coils to the right. The spherical structures in this image represent the various atoms in the sugars and bases (dark gray), and phosphates (light gray). (Courtesy of Kenneth Eward/BioGrafx/Photo Researchers, Inc.)

DNA by the attachment of the phosphate group of one nucleotide to the 3' carbon of a second nucleotide. Since the phosphate of the first nucleotide is attached to its 5' carbon, the DNA chain is said to grow in the 5' to 3' direction. The internal OH group is lost when the two nucleotides are joined together.



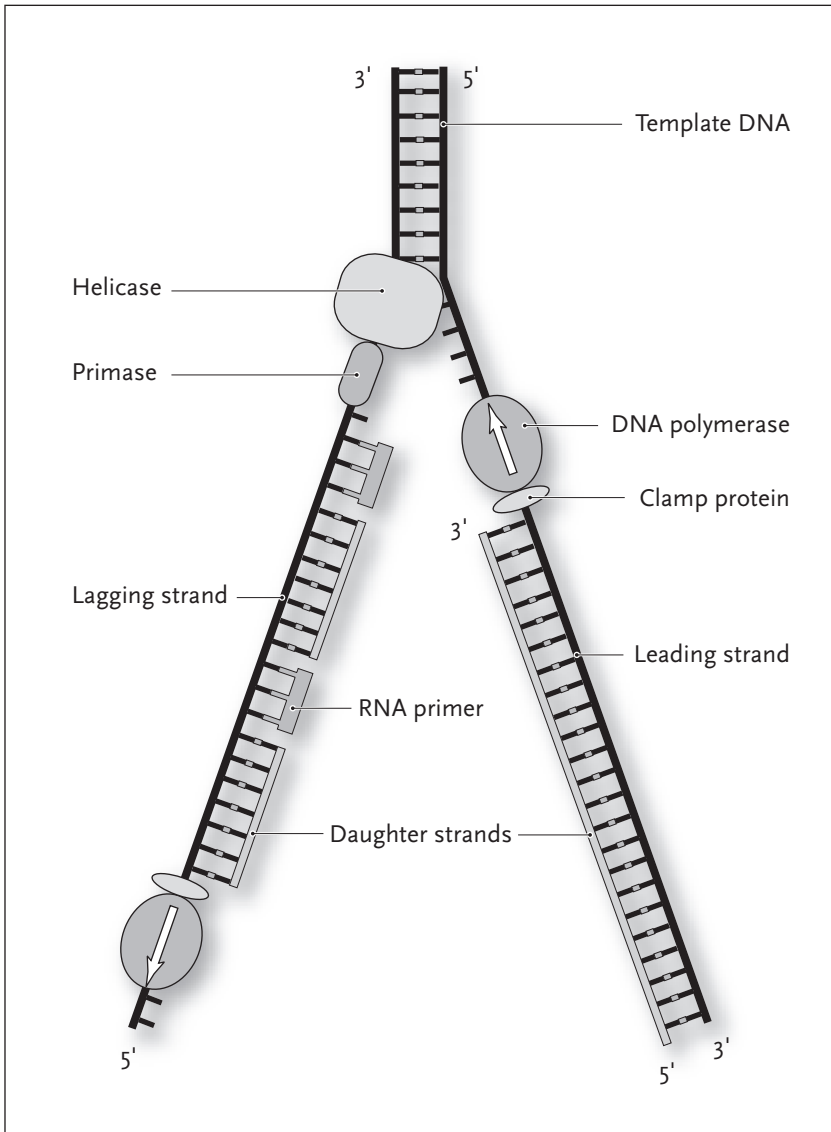
DNA structure. Nucleotides are linked together to form a double-stranded molecule. The bases are held together with double or triple bonds, and the two strands are antiparallel.

Double-stranded DNA forms when two chains of nucleotides interact through the formation of chemical bonds between complementary base pairs. Triple bonds are formed between cytosine and guanine, and double bonds are formed between adenine and thymine. The geometry of each DNA strand is such that they can only form a double-stranded molecule by being antiparallel, that is, the orientation of one strand is in the 5' to 3' direction, while the complementary strand runs 3' to 5'.

In prokaryotes and eukaryotes, duplication of the genome occurs at replication bubbles, each of which contains two replication forks. The bubbles are regions of the DNA that have dissociated so that daughter strands can be synthesized. A prokaryote chromosome has a single replication bubble, whereas a eukaryote chromosome has many. Consequently, DNA replication in eukaryotes does not begin at one end of the chromosome and continue until it reaches the other end; instead, replication begins at many

places simultaneously and progresses until all the bubbles have fused, at which point duplication of the chromosome is complete.

DNA replication requires the coordinated effort of a team of enzymes, led by DNA helicase and primase. The helicase is a remarkable



DNA replication. The helicase separates the two strands so the DNA polymerase can synthesize new strands. The primase provides replication signals for the polymerase, in the form of RNA primers, and the clamp protein keeps the polymerase from falling off the DNA. The leading strand requires only a single primer (not shown). The lagging strand requires many primers, and the daughter strand is synthesized as a series of DNA fragments that are later joined into one continuous strand.

enzyme that is responsible for initiating the formation of the bubble and for separating the two DNA strands. This enzyme moves at an astonishing rate, separating 1,000 nucleotides every second. Helicase gets its name from the fact that DNA is a helix, wherein the two strands coil around each other like two strands in a piece of rope. Consequently, helicase is not just separating the two strands but also unwinding them as the replication fork progresses.

The enzyme that is directly responsible for reading the template strand, and for constructing the new daughter strand, is called DNA polymerase. This enzyme reads the parental DNA in the 3' to 5' direction and creates a daughter strand that grows 5' to 3'. DNA polymerase also has an editorial function: It checks the preceding nucleotide to make sure it is correct before it will add a nucleotide to the growing chain. The editor function of this enzyme introduces an interesting problem: How can the polymerase add the very first nucleotide, given that it has to check a preceding nucleotide before adding a new one? A special enzyme called primase, which is attached to the helicase, solves this problem. Primase synthesizes short pieces of RNA that form a DNA-RNA double-stranded region. The RNA becomes a temporary part of the daughter strand, thus priming the DNA polymerase by providing the crucial first nucleotide in the new strand. Once the chromosome is duplicated, DNA repair enzymes remove the RNA primers and replace them with DNA nucleotides.

For every replication fork there is a leading and lagging strand. The leading strand reads 3' to 5' in the direction of the fork, whereas the lagging strand reads 5' to 3'. Since DNA polymerase reads the template 3' to 5', the leading strand only requires a single primer, which is laid down as soon as the replication bubble forms. The lagging strand, however, has to have many primers, and these are laid down as the fork progresses. This is why the helicase and the primase, as shown in the figure on page 65, are located on the lagging strand. DNA polymerase can duplicate the leading strand continuously, in one long piece, whereas the lagging strand has to be duplicated with many discontinuous pieces, from one primer to the next.

This may seem like an awkward solution, but the alternative is to place a single primer at the 3' tips of each strand and then replicate them both continuously. But this would mean that one helicase and one

polymerase would have to travel the entire length of each strand. Even with the helicase moving as fast as it does, it would be a very slow way of replicating the genome. Using multiple replication forks per chromosome is analogous to parallel processing, which can dramatically reduce the time it takes to complete a task.

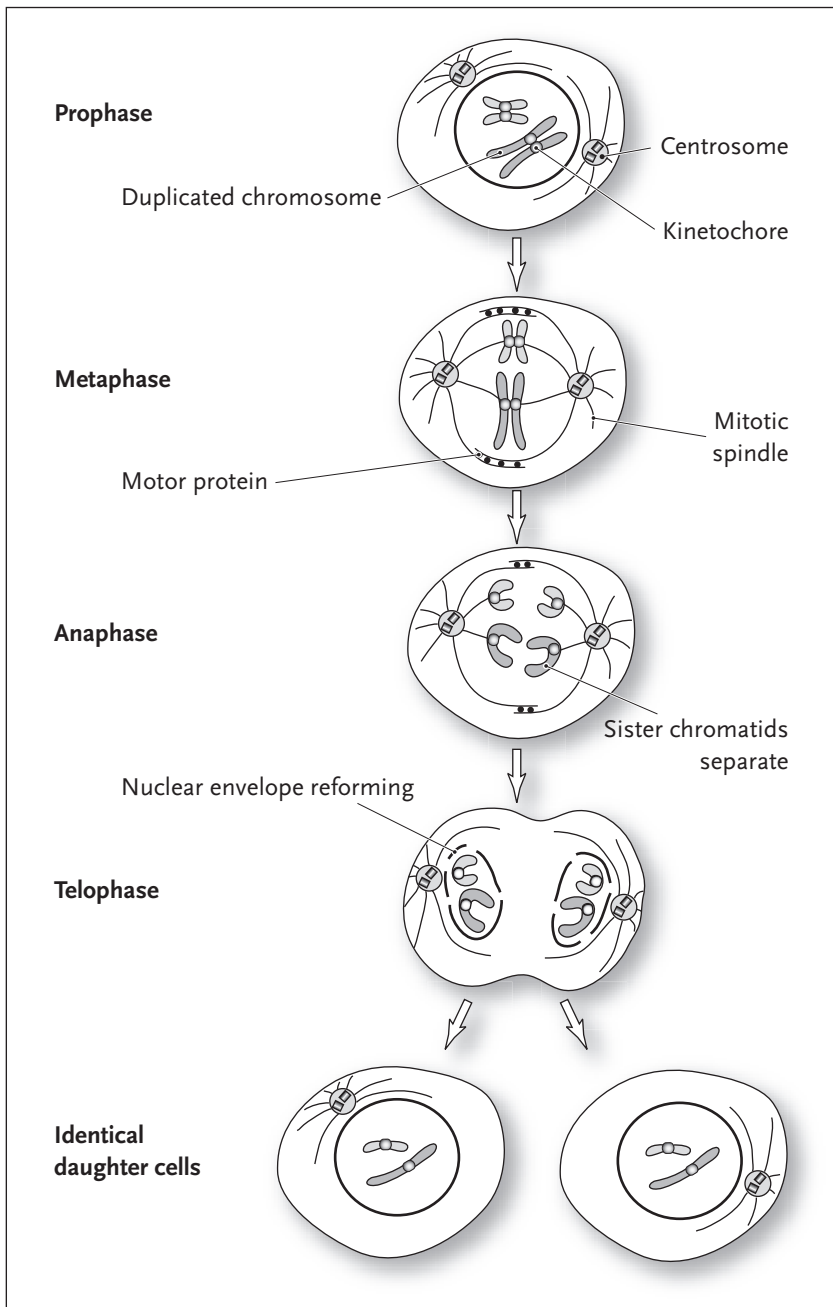
Binary Fission

Prokaryotes all divide by a simple process called binary fission. These cells grow continuously during interphase and the growth is coordinated with the duplication of the chromosome. Soon after the replication bubble forms, one strand of the chromosome is attached to the membrane at one end of the cell, and the other strand is attached to the opposite end. Growth of the cell and the completion of DNA replication are coordinated with the formation of a septum, or invagination, of the cell membrane, which eventually divides the cell into two. The formation of the septum is initiated by a protein called FtsZ, but the details of the process are still unclear.

Mitosis

Cell division among the eukaryotes is more complicated than it is in prokaryotes. The biggest problem eukaryotes have to deal with, compared with prokaryotes, is their enormous genome, spread out over many chromosomes, which must be replicated, packaged, arranged, and sorted before the cell can divide. Division is by a process known as mitosis, which is divided into four stages: prophase, metaphase, anaphase, and telophase. All of these stages are marked out in accordance with the behavior of the nucleus and the chromosomes.

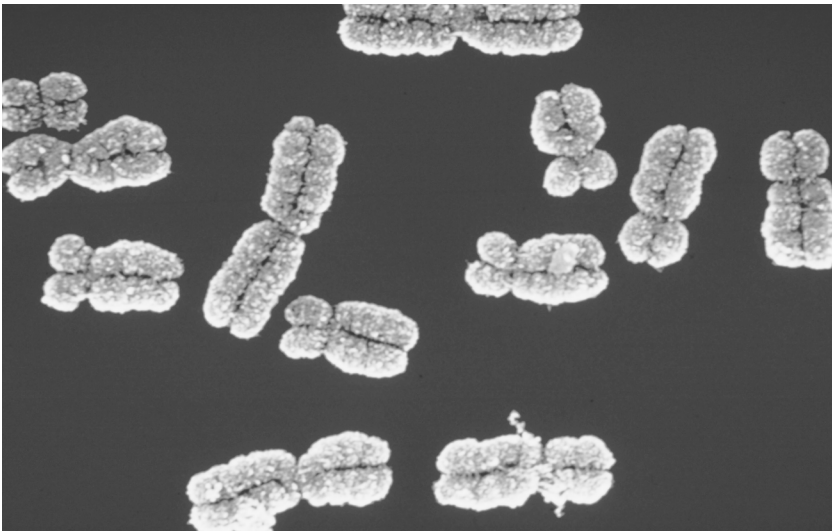
Prophase marks the period during which the duplicated chromosomes begin condensation, and the two centrosomes begin moving to opposite poles of the cell. Under the microscope, the chromosomes become visible as X-shaped structures, which are the two duplicated chromosomes, often called sister chromatids. A special region of each chromosome, called a centromere, holds the chromatids together. Proteins bind to the centromere to form a structure called the kinetochore (see the figure on page 68).



Mitosis. Principal stages dealing with the movement and partitioning of the chromosomes between the future daughter cells. For clarity, only two chromosomes are shown.

Metaphase is a period during which the chromosomes are sorted out and aligned between the two centrosomes. By this time, the nuclear membrane has completely broken down. The two centrosomes and the microtubules fanning out between them form the mitotic spindle. The area in between the spindles, where the chromosomes are aligned, is often referred to as the metaphase plate. Some of the microtubules make contact with the kinetochores, while others overlap, with motor proteins situated in between (for clarity, only two chromosomes are shown in the figure on page 68). Eukaryotes are normally diploid, so a cell would have two copies of each chromosome, one from the mother and one from the father.

Anaphase is characterized by the movement of the duplicated chromosomes to opposite poles of the cell. The first step is the release of an enzyme that breaks the bonds holding the kinetochores together, thus allowing the sister chromatids to separate from each other while remaining bound to their respective microtubules. Motor proteins then move along the microtubule dragging the chromosomes to opposite ends of the cell. Using energy supplied by ATP, the motor proteins break the microtubule down as it drags the chromosome along so that



Scanning electron micrograph of human chromosomes in metaphase.
(Courtesy of Biophoto Associates/Photo Researchers, Inc.)

the microtubule is gone by the time the chromosome reaches the spindle pole. Throughout this process, the motor proteins and the chromosome manage to stay one step ahead of the disintegrating microtubule. The overlapping microtubules aid movement of the chromosomes toward the poles as another type of motor protein pushes the microtubules in opposite directions effectively forcing the centrosomes towards the poles. This accounts for the greater overlap of microtubules in metaphase as compared with anaphase.

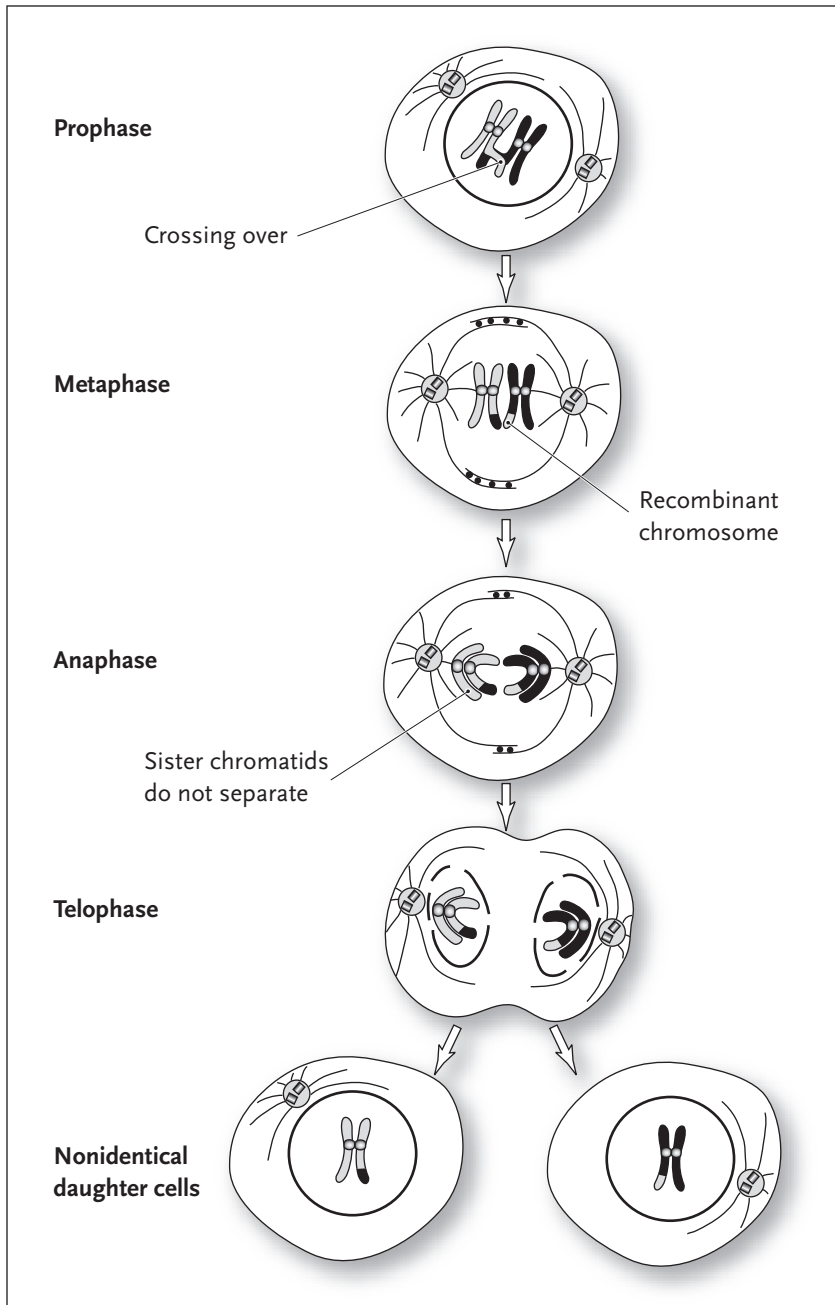
During telophase, the daughter chromosomes arrive at the spindle poles and decondense to form the relaxed chromatin characteristic of interphase nuclei. The nuclear envelope begins forming around the chromosomes, marking the end of mitosis. During the same period, a contractile ring, made of the proteins myosin and actin, begins pinching the parental cell in two. This stage, separate from mitosis, is called cytokinesis and leads to the formation of two daughter cells, each with one nucleus.

Meiosis

Unlike mitosis, which leads to the growth of an organism, meiosis is intended for sexual reproduction and occurs exclusively in ovaries and testes. Eukaryotes, being diploid, receive chromosomes from both parents; if gametes were produced using mitosis, a catastrophic growth in the number of chromosomes would occur each time a sperm fertilized an egg. Meiosis is a special form of cell division that produces haploid gametes (eggs and sperm), each possessing half as many chromosomes as the diploid cell. When haploid gametes fuse, they produce an embryo with the correct number of chromosomes.

The existence of meiosis was first suggested 100 years ago when microbiologists counted the number of chromosomes in somatic and germ cells. The roundworm, for example, was found to have four chromosomes in its somatic cells but only two in its gametes. Many other studies also compared the amount of DNA in nuclei from somatic cells and gonads, always with the same result: The amount of DNA in somatic cells is exactly double the amount in fully mature gametes.

To understand how this could be, scientists studied cell division in the gonads and were able to show that meiosis occurs as two rounds of cell



Meiosis I. The most notable features include genetic recombination (crossing over) between the homologous chromosomes during prophase, comigration of the sister chromatids during anaphase, and the production of nonidentical daughter cells. Only one homologous pair is shown.

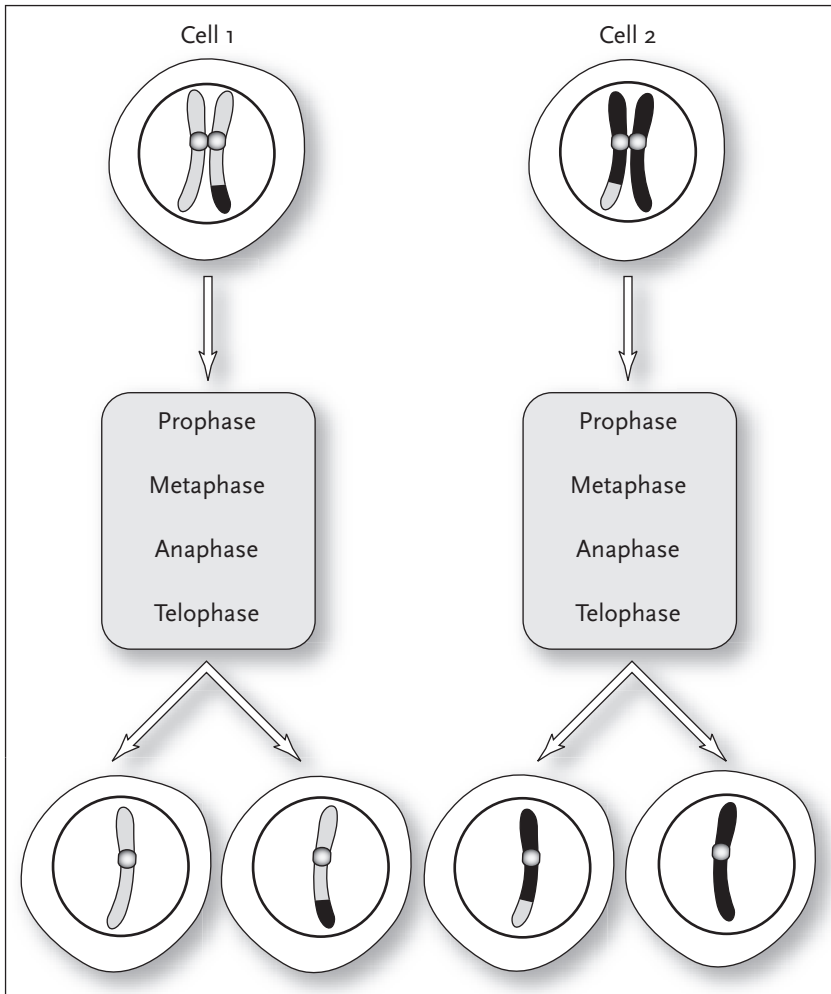
division with only one round of DNA synthesis. The two rounds of division were called meiosis I and meiosis II, and scientists observed that both could be divided into the same four stages known to occur in mitosis. Indeed, meiosis II is virtually identical to a mitotic division. Meiosis I resembles mitosis, but close examination shows three important differences: gene swapping occurs between homologous chromosomes in prophase; homologs (i.e., two homologous chromosomes) remain paired at metaphase, instead of lining up at the plate as is done in mitosis; and the kinetochores do not separate at anaphase.

Homologous chromosomes are two identical chromosomes that come from different parents. For example, humans have 23 chromosomes from the father and the same 23 from the mother. We each have a maternal chromosome 1 and a paternal chromosome 1; they carry the same genes but specify slightly different traits. Chromosome 1 may carry the gene for eye color, but the maternal version, or allele, may specify blue eyes, whereas the paternal allele specifies brown. During prophase, homologous pairs exchange large numbers of genes by swapping whole pieces of chromosome. Thus one of the maternal chromatids (gray in the figure on page 71) ends up with a piece of paternal chromosome, and a paternal chromatid receives the corresponding piece of maternal chromosome. Mixing genetic material in this way is unique to meiosis, and it is one of the reasons sexual reproduction has been such a powerful evolutionary force.

During anaphase of meiosis I, the kinetochores do not separate as they do in mitosis. The effect of this is to separate the maternal and paternal chromosomes by sending them to different daughter cells, although the segregation is random. That is, the daughter cells receive a random assortment of maternal and paternal chromosomes, rather than one daughter cell receiving all paternal chromosomes and the other all maternal chromosomes. Random segregation, along with genetic recombination, accounts for the fact that while children resemble their parents, they do not look or act exactly like them. These two mechanisms are responsible for the remarkable adaptability of all eukaryotes.

Meiosis II begins immediately after the completion of meiosis I, which produces two daughter cells, each containing a duplicated parental chromosome and a recombinant chromosome consisting of

both paternal and maternal DNA. These two cells divide mitotically to produce four haploid cells, each of which is genetically unique, containing unaltered or recombinant maternal and paternal chromosomes. This example follows the fate of a single homologous pair. In reality, the four haploid cells, if human, would contain 23 chromosomes each,



Meiosis II. Cells 1 and 2 were produced in meiosis I. These cells divide mitotically, but without replication of the chromosomes, to produce four genetically unique haploid gametes.

some from the mother, some from the father, some remixed (recombinant), others unaltered.

Meiosis produces haploid cells by passing through two rounds of cell division with only one round of DNA synthesis. However, as we have seen, the process is not just concerned with reducing the number of chromosomes but is also involved in stirring up the genetic pot in order to produce unique gametes that may someday give rise to an equally unique individual.

.5. GENES

Genes are a precious biological commodity: They hold the key to a cell's survival and are the summation of countless natural experiments over millions of years of evolution. Nature has poked, prodded, modified, and tweaked each gene, almost to exhaustion. Those that fail the test, or do not live up to expectations, are eliminated. The genes that we find today, in ourselves and in other organisms, are the survivors. The elaborate codes they carry, and the proteins or RNA molecules they specify, produce organisms that are also survivors: hardy creatures that are resourceful, diligent, and well adapted to their environment.

The attention nature pays to a gene is analogous to the labors of an artist, who spends many hours carefully shaping, carving, and polishing a sculpture. Work on the sculpture may last for weeks or years and often goes through many revisions, each of which brings the object closer to the artist's vision. The difference between the artist and nature is that nature works on its sculptures indirectly, by polishing or altering genes; removing a base here, adding one there, and seeing what happens. In many cases, nature is never satisfied with the outcome, and so the gene, and its product, continues changing over time. Sometimes, however, nature seems to hit it right, and the gene remains unchanged for millions of years.

These and many other insights into the nature of the gene have been hard-won, over many years, by thousands of researchers studying different species of plant and animal life. But the pace of this kind of work increased dramatically throughout the late 1990s with the initiation of the human genome project. The final draft of the human genome was

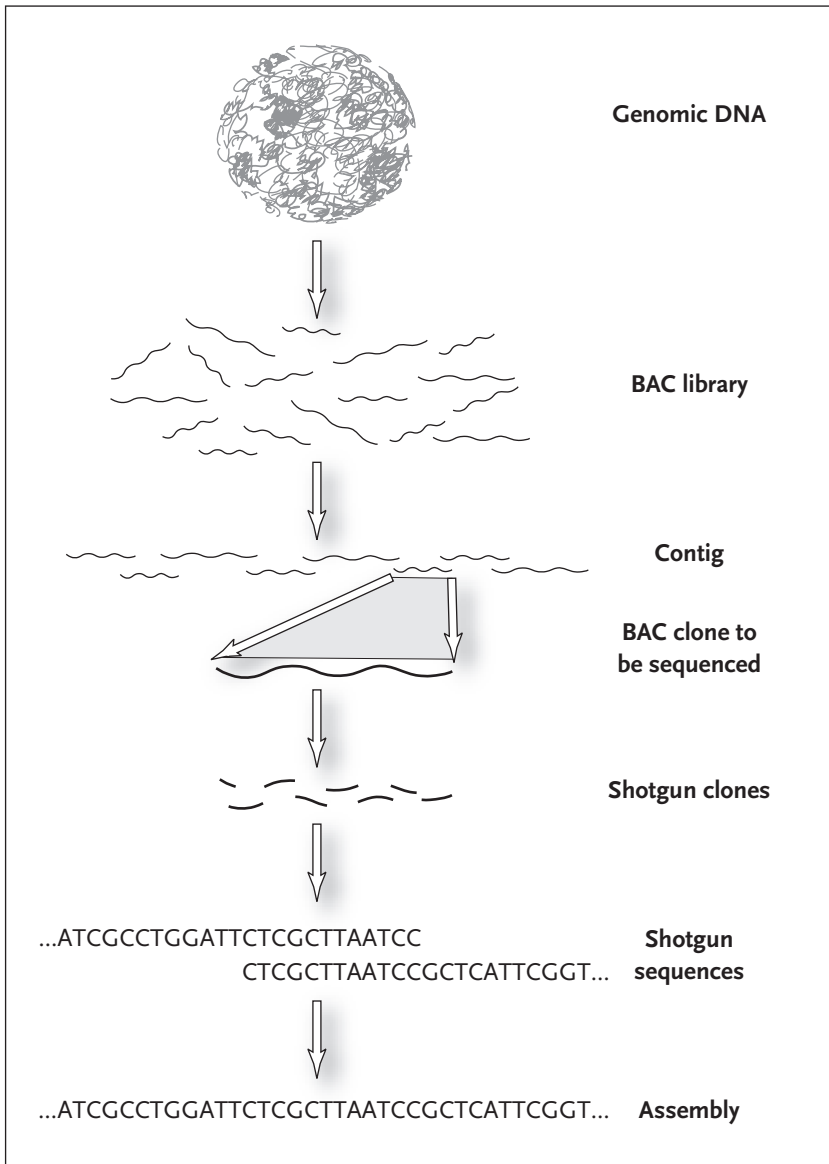
completed in 2003 and provides a wealth of information regarding the structure, organization, and evolution of our genes.

The Human Genome Project

Sequencing the entire human genome is an idea that grew over a period of 20 years, beginning in the early 1980s. At that time, the DNA sequencing method invented by the British biochemist Fred Sanger, then at the University of Cambridge, was but a few years old and had only been used to sequence viral or mitochondrial genomes (see chapter 8 for a description of sequencing methods). Indeed, one of the first genomes to be sequenced was that of bacteriophage G4, a virus that infects the bacterium *Escherichia coli* (*E. coli*). The G4 genome consists of 5,577 nucleotide pairs (or base pairs, abbreviated bp) and was sequenced in Dr. Sanger's laboratory in 1979. By 1982, the Sanger protocol was used by others to sequence the genome of the animal virus SV40 (5,224 bp), the human mitochondrion (16,569 bp), and bacteriophage lambda (48,502 bp). Besides providing invaluable data, these projects demonstrated the feasibility of sequencing very large genomes.

The possibility of sequencing the entire human genome was first discussed at scientific meetings organized by the United States Department of Energy (DOE) between 1984 and 1986. A committee appointed by the U.S. National Research Council endorsed the idea in 1988 but recommended a broader program to include the sequencing of the genes of humans, bacteria, yeast, worms, flies, and mice. They also called for the establishment of research programs devoted to the ethical, legal, and social issues raised by human genome research. The program was formally launched in late 1990 as a consortium consisting of coordinated sequencing projects in the United States, Britain, France, Germany, Japan, and China. At about the same time, the Human Genome Organization (HUGO) was founded to provide a forum for international coordination of genomic research.

By 1995, the consortium had established a strategy, called hierarchical shotgun sequencing, which they applied to the human genome as well as to the other organisms mentioned. With this strategy, genomic DNA is cut into one-megabase (Mb) fragments (i.e., each fragment consists of 1 million bases) that are cloned into bacterial artificial



Hierarchical shotgun sequencing. Total genomic DNA is cut with a restriction enzyme into one-megabase fragments (i.e., 1 million base pairs per fragment) that are cloned into bacterial artificial chromosomes (BACs) to form a library. The BAC fragments are partially characterized in order to organize them into an overlapping assembly called a contig. Clones are selected from the contigs for shotgun sequencing and final assembly.

chromosomes (BACs) to form a library of DNA fragments. The BAC fragments are partially characterized, then organized into an overlapping assembly called a contig. Clones are selected from the contigs for shotgun sequencing. That is, each shotgun clone is digested into small 1,000 bp fragments, sequenced, and then assembled into the final sequence with the aid of computers. Organizing the initial BAC fragments into contigs greatly simplifies the final assembly stage.

Sequencing of the human genome was divided into two stages. The first stage, completed in 2001, was a rough draft that covered about 80 percent of the genome with an estimated size of more than 3 billion bases (also expressed as 3 gigabases, or 3 Gb). The final draft, completed in April 2003, covers the entire genome and refines the data for areas of the genome that were difficult to sequence. It also filled in many gaps that occurred in the rough draft. The final draft of the human genome gives us a great deal of information that may be divided into three categories: gene content, gene origins, and gene organization.

GENE CONTENT Analysis of the final draft has shown that the human genome consists of 3.2 Gb of DNA, that encodes about 30,000 genes (estimates range between 25,000 to 32,000). The estimated number of genes is surprisingly low; many scientists had believed the human genome contained 100,000 genes. By comparison, the fruit fly has 13,338 genes, and the simple roundworm, *Caenorhabditis elegans* (*C. elegans*), has 18,266. The genome data suggests that human complexity, as compared to the fruit fly or the worm, is not simply due to the absolute number of genes but also involves the complexity of the proteins that are encoded by those genes. In general, human proteins tend to be much more complex than those of lower organisms. Data from the final draft and other sources provides a detailed overview of the functional profile of human cellular proteins (see table on page 79).

GENE ORIGINS Fully one half of human genes originated as transposable elements, also known as jumping genes (these will be discussed at length in a following section). Equally surprising is the fact that 220 of our genes were obtained by horizontal transfer from bacteria, rather than ancestral, or vertical, inheritance. In other words, we obtained these genes directly from bacteria, probably during

FUNCTIONAL PROFILE OF KNOWN CELLULAR PROTEINS

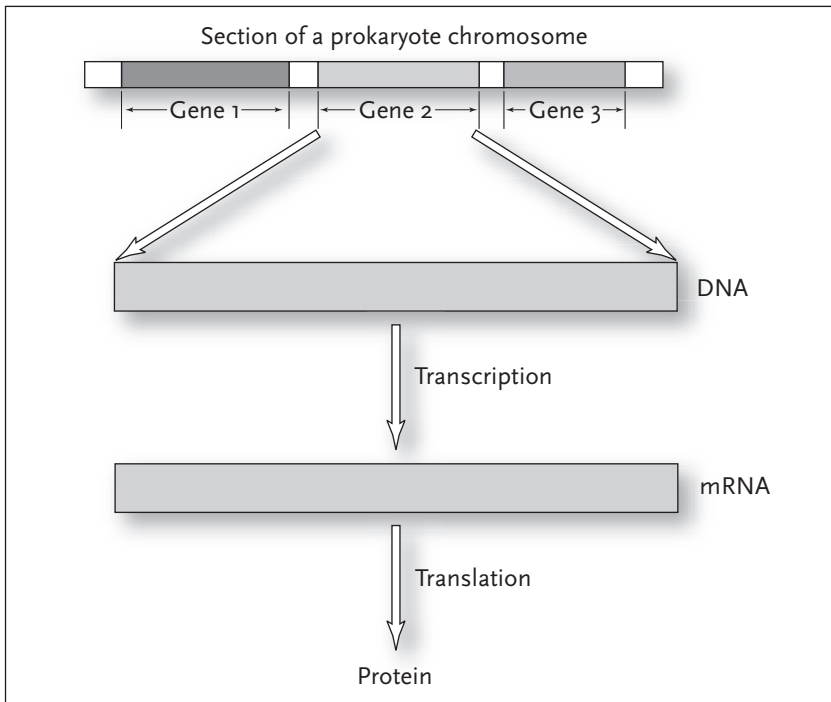
Process	Number of Proteins
Energy metabolism	5,200
DNA replication/repair	900
Transcription/translation	3,200
Signaling (intra- and extra-cellular)	3,100
Protein modifiers	850
Transport	1,200
Multifunctional proteins	400
Structural	900
Defense	1,050
Total	16,800*

*The total shown is about half of the estimated 32,000 protein encoding genes in the human genome. The discrepancy is due to the fact that many genes have an unknown function.

episodes of infection, in a kind of natural gene therapy, or gene swapping. We know this to be the case because, while these genes occur in bacteria, they are not present in yeast, fruit flies, or any other eukaryotes that have been tested.

The function of most of the horizontally transferred genes is unclear, although a few may code for basic metabolic enzymes. A notable exception is a gene that codes for an enzyme called monoamine oxidase (MAO). Monoamines are neurotransmitters, such as dopamine, norepinephrine, and serotonin, which are needed for neural signaling in the human central nervous system. Monoamine oxidase plays a crucial role in the turnover of these neurotransmitters (see chapter 6). How MAO, obtained from bacteria, could have developed such an important role in human physiology is a great mystery.

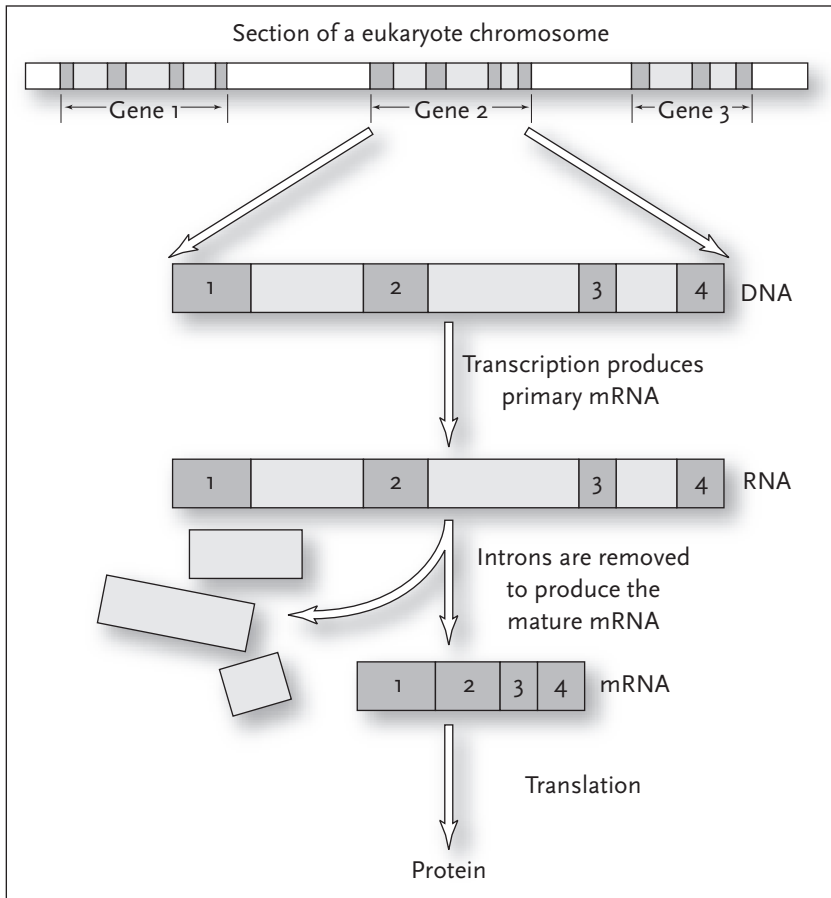
GENE ORGANIZATION In prokaryotes, genes are simply arranged in tandem along the chromosome, with little if any DNA separating one gene from the other. Each gene is transcribed into messenger RNA (mRNA), which is translated into protein. Indeed, in prokaryotes, which



Prokaryote genes. The genes are arranged in tandem along the chromosome with little, if any, DNA separating one gene from the other. The genes may code for protein, as shown above for Gene 2, or ribosomal RNA (rRNA).

have no nucleus, translation often begins even before transcription is complete. In eukaryotes, as we might expect, gene organization is more complex. Data from the genome project shows clearly that eukaryote genes are split into subunits, called exons, and that each exon is separated by a length of DNA, called an intron. A gene, consisting of introns and exons, is separated from other genes by long stretches of noncoding DNA called intervening sequences. Eukaryote genes are transcribed into a primary RNA molecule that includes exon and intron sequences. The primary transcript never leaves the nucleus and is never translated into protein. Nuclear enzymes remove the introns from the primary transcript, after which the exons are joined together to form the mature mRNA. Thus only the exons carry the necessary code to produce a protein.

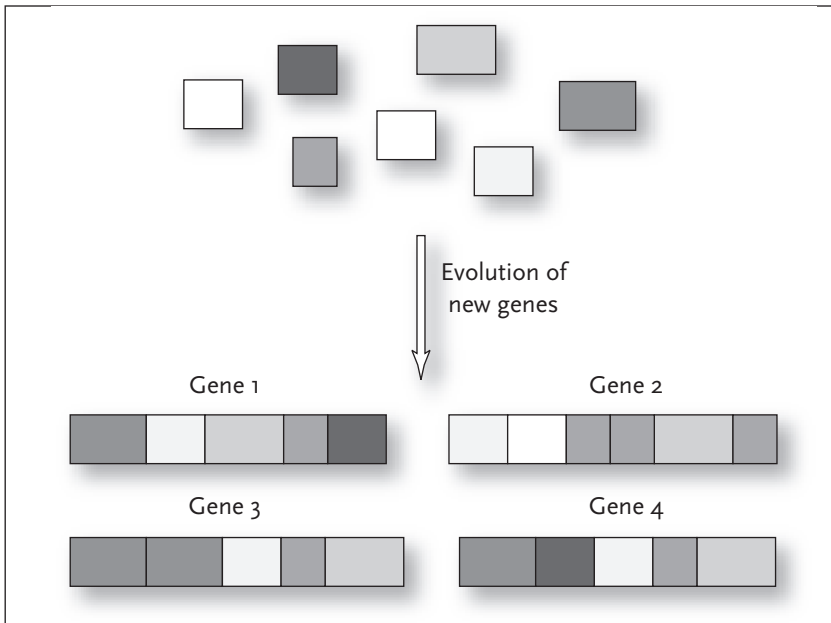
Why do eukaryotes, all of which have split genes, go through such a laborious procedure when the simple and direct method used by prokaryotes works so well and seems so eloquent? The answer lies with



Eukaryote genes. The genes are arranged in separate subunits called exons (dark gray) and introns (light gray). Each gene, which consists of introns and exons, is separated from other genes by long stretches of noncoding DNA called intervening sequences (white areas). The genes are transcribed into a primary RNA molecule that includes exon and intron sequences. Nuclear enzymes remove the introns from the primary transcript, and then the exons are joined together to form the mature mRNA, which is translated into protein. Transcription of ribosomal genes is similar, except that the exons become the individual rRNAs.

the need to maximize protein evolution and to protect the genome from insertional mutagenesis. The rate at which proteins evolve can be maximized by a process known as exon shuffling. Any given protein usually has more than one job that it is good at. One end of a protein may be good at binding to a specific region of the genome, while the other end specializes in phosphorylating other proteins. The middle region of the protein may give it a special corkscrew shape. If each of these regions is encoded by three different exons, then it is easy to see how new proteins could emerge very quickly by recombining, or shuffling, preexisting exons.

Insertional mutagenesis occurs because our world is full of viruses that can infect eukaryote cells. The life cycle of many of these viruses is such that they are able to insert their genome into a chromosome of the



Exon shuffling. New eukaryote genes can evolve without the appearance of new mutations simply by recombining, or shuffling, preexisting exons. In the example shown, seven exons were used to produce four novel genes, each of which would code for a novel protein. Exon shuffling is not possible in a prokaryote genome.

host cell. If eukaryotes had a simple prokaryote-like genome, then no matter where the virus inserted, it would likely destroy, damage, or mutate a gene. On the other hand, the eukaryote genome contains large regions of noncoding DNA (introns and intervening sequences) so that, in all probability, insertion of a virus will not damage any of the genes. The only way a virus can damage any of our genes is if it happens to insert at an exon-intron boundary. If this happens, special DNA sequences located at the boundary are corrupted, leading to the production of defective messenger RNA.

The fact is that human genes and the genes of most eukaryotes are hidden in a sea of noncoding DNA. As already mentioned, the human genome consists of 3.2 Gb of DNA and contains about 30,000 genes. With an average gene size of 1,200 bp (based on exons only), this amounts to a total of 36 Mb of coding DNA, or roughly 1.0 percent of the total DNA content. The vast majority of our DNA codes for nothing but has a crucial role in protecting our genes from marauding pieces of DNA.

The Future of Genomic Research

The many insights into the structure, evolution, and organization of our genes that the genome project has provided are expected to revolutionize the study of many medical disorders. Various research communities, as diverse as nursing, psychiatry, and gene therapy, have already written summary articles outlining the many advances they expect to make with the available sequence data. Cancer research provides the most striking example of the impact this data has already had. Scientists at the Sanger Institute in Cambridge, England, established a research program in 2001 to use human sequence data to identify all cancer-causing genes in the human genome. Within a few months they were able to isolate and fully characterize a gene that causes more than 70 percent of all malignant melanomas. Their analysis is so complete and so illuminating that they believe a cure for this deadly form of cancer will be available within a few years, and they expect to complete the identification of all other cancer genes by 2005. Cancer is only one of many prominent genetic diseases. Similar advances are expected in the discovery and characterization of genes